

Shinbrot Targeting Walkthrough

This Live Script is the principal explanatory walkthrough for the final Shinbrot Lorenz targeting implementation in this folder. It is not a replacement for the verified computational functions. Instead, it calls those functions in a documented order and displays the intermediate diagnostics, tables and figures inline.

The study has three connected parts:

1. a deterministic single-source demonstration comparing the verified Shinbrot discontinuity-aware bisection search with the deterministic grid-first hybrid extension;
2. a deterministic 60-source benchmark loaded from verified saved results generated by the current code;
3. a fixed-step RK4 Gaussian-noise investigation using the verified bisection method with accepted-section retargeting.

The hybrid method is a deterministic extension for comparison. It is not used in the noise investigation.

Reproducible Project Setup

The script first locates the Shinbrot project folder, moves MATLAB into that folder, and adds only that folder to the path. This avoids hidden dependencies on a previous MATLAB session. The switches below prevent expensive studies from being regenerated unless explicitly requested. In normal use the Live Script loads the verified saved ensemble and noise results and then plots them inline.

```
clearvars;
close all;
clc;

projectPath = walkthrough_project_path();
restoredefaultpath;
rehash toolboxcache;
cd(projectPath);
addpath(projectPath);
rng(7, 'twister');

runExpensiveEnsemble = false;
runExpensiveNoiseSweep = true;
runNoiseSmokeOnly = true;
runControlBoundSweep = false;

setupTable = table(string(projectPath), runExpensiveEnsemble, ...
    runExpensiveNoiseSweep, runNoiseSmokeOnly, runControlBoundSweep, ...
    'VariableNames', {'ProjectPath', 'RunExpensiveEnsemble', ...
    'RunExpensiveNoiseSweep', 'RunNoiseSmokeOnly', ...
    'RunControlBoundSweep'});
disp(setupTable);
```

ProjectPath	RunExpensiveEnsemble	RunExpensiveNoiseSweep	RunNoiseSmokeOnly	RunControlBoundSweep
"C:\Users\npcoz\OneDrive - Imperial College London\Project\Codes\Shinbrot targeting"	false	true	true	false

Lorenz System

The controlled Lorenz equations used throughout the implementation are

$$\dot{x} = \sigma(y - x),$$

$$\dot{y} = x(\rho - z) - y + p_{\text{control}},$$

$$\dot{z} = xy - \beta z.$$

The standard Lorenz parameters are $\sigma = 10$, $\rho = 28$ and $\beta = 8/3$. The targeting parameter p_{control} is a bounded additive perturbation in the y -equation. It is not a state kick. During a deterministic targeting replay the selected value is held constant from the source Poincare crossing until the selected target horizon has been reached.

```
verifiedBisectionIdentifier = verified_bisection_identifier();

params.sigma = 10;
params.rho = 28;
params.beta = 8/3;
params.zSection = 26.921;
params.xSectionMin = 8.0;
params.crossingDirection = +1;
params.odeOptions = odeset('RelTol', 1e-9, 'AbsTol', 1e-11);
params.eventDisableTime = 1e-6;
params.maxStepOffSection = 1e-5;
params.sectionTolerance = 1e-9;
params.maxTimeToNextValidSection = 100;
params.maxRejectedCrossings = 100;

lorenzParameterTable = table(params.sigma, params.rho, params.beta, ...
    'VariableNames', {'sigma', 'rho', 'beta'});
disp(lorenzParameterTable);
```

sigma	rho	beta
10	28	2.6667

Poincare Section

The implementation converts the continuous Lorenz flow into a scalar return-map problem by recording accepted crossings of the half-plane

$$z = 26.921, \quad x > 8.$$

The event function detects the directed crossing of the plane $z = 26.921$. The half-plane condition $x > 8$ is then enforced by `next\_valid\_section\_crossing`, which rejects plane crossings on the other side and continues integration. The accepted crossing direction is positive and the section tolerance is `params.sectionTolerance`.

```
sectionTable = table(params.zSection, params.xSectionMin, ...
    params.crossingDirection, params.sectionTolerance, ...
    params.eventDisableTime, params.maxStepOffSection, ...
    'VariableNames', {'zSection', 'xSectionMin', ...
    'CrossingDirection', 'SectionTolerance', 'EventDisableTime', ...
    'MaxStepOffSection'});
disp(sectionTable);
```

zSection	xSectionMin	CrossingDirection	SectionTolerance	EventDisableTime	MaxStepOffSection
26.921	8	1	1e-09	1e-06	1e-05

Target Periodic Orbit

The target is represented on the accepted Poincare section by the fixed scalar coordinate $X_t = 13.729$ and the associated three-dimensional section state. Success is not exact equality with the target point; instead, a replay is successful when the accepted crossing satisfies

$$|X_n - X_t| \leq 0.008.$$

This neighbourhood definition is necessary because the targeting computation is numerical and because the target point is a representative fixed point of the return-map dynamics.

```
target.x = 13.729;
target.state = [13.729; 19.585; params.zSection];
target.tolerance = 0.008;

targetTable = table(target.x, target.state(1), target.state(2), ...
    target.state(3), target.tolerance, ...
    'VariableNames', {'TargetX', 'TargetStateX', 'TargetStateY', ...
    'TargetStateZ', 'TargetTolerance'});
disp(targetTable);
```

TargetX	TargetStateX	TargetStateY	TargetStateZ	TargetTolerance
13.729	13.729	19.585	26.921	0.008

Search Settings

Both deterministic methods use the same dynamical settings, target tolerance and parameter bound. The admissible parameter interval is $[-\Delta p, \Delta p]$ with $\Delta p = 0.1$. The deterministic search horizon is the number of accepted Poincare crossings after the source state. The hybrid method additionally evaluates the full parameter grid before refining verified candidates.

```
control.deltaP = 0.1;
control.maxSearchCrossings = 12;
control.numParameterSamples = 81;
control.bisectionIterations = 24;
control.maxDiscontinuityIsolationDepth = 24;
control.maxDiscontinuityIntervals = 256;
control.searchMethod = 'hybridGridBisection';

benchmarkSettings.repetitions = 5;

controlTable = table(control.deltaP, control.maxSearchCrossings, ...
    control.numParameterSamples, control.bisectionIterations, ...
    control.maxDiscontinuityIsolationDepth, ...
    control.maxDiscontinuityIntervals, ...
    string(verifiedBisectionIdentifier), ...
    'VariableNames', {'DeltaP', 'MaxSearchCrossings', ...
    'ParameterSamples', 'BisectionIterations', ...
    'MaxDiscontinuityIsolationDepth', 'MaxDiscontinuityIntervals', ...
    'VerifiedBisectionIdentifier'});
disp(controlTable);
```

DeltaP	MaxSearchCrossings	ParameterSamples	BisectionIterations	MaxDiscontinuityIsolationDepth	MaxDiscontinuityIntervals	VerifiedBisectionIdentifier
0.1	12	81	24	24	256	"shinbrotPaperBisection"

Source-State Construction

The nominal return map is generated with `pControl = 0`. The principal source is the full three-dimensional state at the 150th accepted section crossing from the initial Lorenz state `[1;1;1]`. The scalar coordinate is used for the return-map search, but the full state is retained for every independent replay.

```
fprintf('Generating the uncontrolled Poincare return map...\n');
```

Generating the uncontrolled Poincare return map...

```
X0 = [1; 1; 1];
numMapCrossings = 1200;
sourceIndex = 150;
mapData = generate_return_map(X0, numMapCrossings, params, 0);
if size(mapData.states, 2) < numMapCrossings
    error('Only %d valid section crossings were generated.', ...
        size(mapData.states, 2));
end

sourceState = mapData.states(:, sourceIndex);
if abs(sourceState(1) - target.x) <= target.tolerance
    error('The configured source point lies inside the target interval.');
```

```
end
sourceX = sourceState(1);

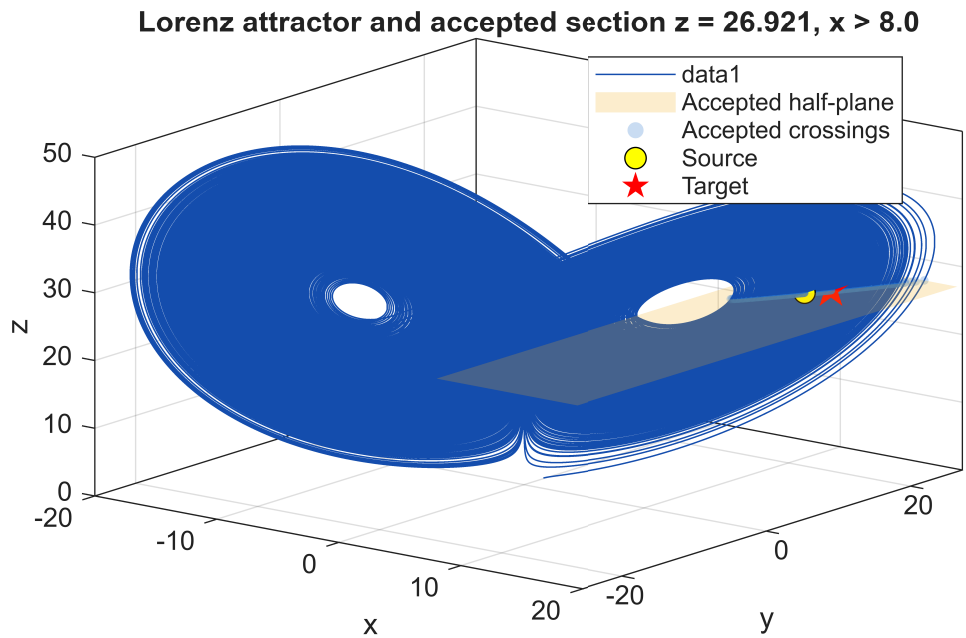
sourceSummary = table(sourceIndex, sourceX, target.x, ...
    abs(sourceX - target.x), numMapCrossings, ...
    'VariableNames', {'SourceIndex', 'SourceX', 'TargetX', ...
        'InitialScalarSeparation', 'GeneratedMapCrossings'});
disp(sourceSummary);
```

SourceIndex	SourceX	TargetX	InitialScalarSeparation	GeneratedMapCrossings
150	12.807	13.729	0.92231	1200

Lorenz Attractor, Section and Accepted Crossings

The following figure shows the nominal trajectory used to build the return map, the accepted half-plane, the target point and the selected source point. The accepted section states are overlaid so the scalar sequence can be seen as samples of the continuous flow.

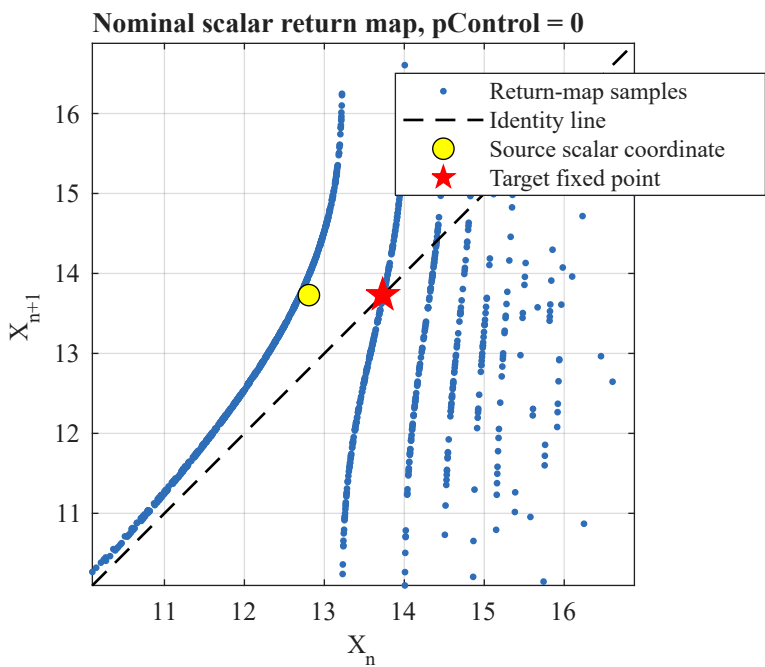
```
walkthrough_plot_lorenz_section(mapData, sourceState, target, params);
```



Scalar Return Map

Consecutive accepted crossings define a one-dimensional return-map view: $X_{n+1} = F(X_n)$. The target fixed point is shown against the identity line. This plot is only a scalar projection of the full section state; all replay verification still reintegrates the full Lorenz equations.

```
walkthrough_plot_return_map(mapData, sourceState, target);
```



Uncontrolled Baseline

The uncontrolled baseline starts from the same source state and sets `pControl = 0`. It counts accepted section crossings until the target neighbourhood is reached or until the configured maximum number of crossings is exhausted. This is the natural-recurrence reference against which the controlled searches are compared.

fprintf('Running the uncontrolled recurrence baseline...\n');																
Running the uncontrolled recurrence baseline...																
uncontrolled = uncontrolled_hitting_time_1d(... sourceState, target, params, 2000);																
uncontrolledTable = table(uncontrolled.hit, uncontrolled.numCrossings, ... uncontrolled.physicalTime, numel(uncontrolled.xSequence), ... 'VariableNames', {'Hit', 'AcceptedCrossingsToHit', ... 'PhysicalTimeToHit', 'StoredScalarSequenceLength'}); disp(uncontrolledTable);																
	Hit	AcceptedCrossingsToHit	PhysicalTimeToHit	StoredScalarSequenceLength												
	true	266	400.01	267												

Shinbrot Discontinuity-Aware Bisection

The verified Shinbrot search calls `search\_parameter\_to\_target` with the dispatcher identifier `shinbrotDiscontinuityAwareBisection`. Internally, `run\_shinbrot\_discontinuity\_aware\_bisection` subdivides the full bounded interval, compares ordered left/right crossing signatures, isolates continuous target brackets, and then refines only brackets whose endpoint images remain on the same branch and contain the target.

This section does not reproduce a second bisection algorithm. It calls the verified `m` implementation and then displays the search metadata: discontinuity counts, crossing-signature evaluations, bracket bounds, selected horizon, selected parameter and target error.

fprintf('Running the verified Shinbrot bisection search...\n');																
Running the verified Shinbrot bisection search...																
bisectionControl = control; bisectionControl.searchMethod = verifiedBisectionIdentifier; bisectionSearch = search_parameter_to_target(... sourceState, target, params, bisectionControl, ... verifiedBisectionIdentifier); assert_verified_bisection_search(... bisectionSearch, verifiedBisectionIdentifier, ... 'principal deterministic bisection');																
bisectionReplayTimer = tic; bisectionReplay = walkthrough_replay_parameter(... sourceState, target, params, bisectionSearch); bisectionReplay.replayTime = toc(bisectionReplayTimer);																
bisectionSearchTable = walkthrough_bisection_diagnostics(... bisectionSearch, bisectionReplay); disp(bisectionSearchTable);																
	Method	SearchFound	ReplayHit	SelectedP	AbsP	Horizon	SearchTargetError	ReplayTargetError	SearchTime	ReplayTime	ParameterEvaluations	CrossingPropagations	TotalBisectionIterations	CrossingSignatureEvaluations	DiscontinuitiesDetected	DiscontinuityReductions
	"shinbrotPaperBisection"	true	true	0.086546	0.086546	3	4.0636e-09	4.0648e-09	1.9503	0.071135	60	347	52	60	29	28

Hybrid Grid-First/Bisection Method

The hybrid method is an addition to the original Shinbrot procedure. At each horizon it evaluates the complete parameter grid, records all direct grid hits and all adjacent sign-change brackets, then bisects each verified adjacent bracket. Candidate selection is lexicographic:

1. smallest targeting horizon; 2. smallest `abs(pControl)`; 3. smallest final target residual.

The search is still verified by direct integration from the original source state. A sign change alone is not treated as a hit.

fprintf('Running the deterministic grid-first hybrid search...\n');																
Running the deterministic grid-first hybrid search...																
hybridControl = control; hybridControl.searchMethod = 'hybridGridBisection'; hybridSearch = search_parameter_to_target(... sourceState, target, params, hybridControl, 'hybridGridBisection');																
hybridReplayTimer = tic; hybridReplay = walkthrough_replay_parameter(... sourceState, target, params, hybridSearch); hybridReplay.replayTime = toc(hybridReplayTimer);																
hybridSearchTable = walkthrough_hybrid_diagnostics(hybridSearch, hybridReplay); disp(hybridSearchTable);																
	Method	SearchFound	ReplayHit	SelectedP	AbsP	Horizon	SearchTargetError	ReplayTargetError	SearchTime	ReplayTime	ParameterEvaluations	CrossingPropagations	GridEvaluations	FinalBracketBisectionIterations	TotalBisectionIterations	BracketLow
	"hybridGridBisection"	true	true	0.019375	0.019375	3	0.00056305	0.00056305	20.18	0.048091	461	1116	243	2	218	0.0175
																BracketHigh
																"grid-firs

```
hybridFeatureTable = walkthrough_hybrid_feature_table( ...
    hybridSearch, target);
disp(hybridFeatureTable);
```

Horizon	DirectGridHits	AdjacentBrackets	VerifiedCandidates	MinimumGridDistance	TargetTolerance
1	0	0	0	0.22626	0.008
2	0	1	0	1.2759	0.008
3	0	14	7	0.010497	0.008
4	0	0	0	NaN	0.008
5	0	0	0	NaN	0.008
6	0	0	0	NaN	0.008
7	0	0	0	NaN	0.008
8	0	0	0	NaN	0.008
9	0	0	0	NaN	0.008
10	0	0	0	NaN	0.008
11	0	0	0	NaN	0.008
12	0	0	0	NaN	0.008

Principal Deterministic Comparison

Both procedures are now compared from the same source. The reference values below are verification references from the current implementation, not values used to select the controls. The table is generated from the executed searches and independent replays.

```
principalComparison = walkthrough_principal_comparison_table( ...
    bisectionSearch, bisectionReplay, hybridSearch, hybridReplay);
disp(principalComparison);
```

Method	Success	ReplayHit	SelectedControl	AbsControl	Horizon	FinalTargetError	ReplayTargetError	SearchTime	ReplayTime	TotalComputationalTime	SelectionMethod
"Shinbrot bisection"	true	true	0.086546	0.086546	3	4.0636e-09	4.0648e-09	1.9503	0.071135	2.0215	"Shinbrot paper bisection with crossing-order discontinuity reduction"
"Grid-first hybrid"	true	true	0.019375	0.019375	3	0.00056305	0.00056305	20.18	0.048091	20.228	"grid-first verified adjacent-grid bracket bisection"

```
% The checks below assert the properties a valid result must have, not the
% particular numbers a given implementation happens to return.  Pinning
% literal values here made the walkthrough a record of one implementation
% rather than a verification of the method, and it is what allowed this
% file and Table 2 of the report to drift apart.  The returned values are
% displayed in principalComparison above rather than asserted.
```

```
walkthrough_assert_valid_result(bisectionSearch, bisectionReplay, ...
    target, control, 'Shinbrot bisection');
```

Shinbrot bisection: verified. p = 0.08654567227, n = 3, replay error = 4.06482e-09, search-replay divergence = 1.21e-12

```
walkthrough_assert_valid_result(hybridSearch, hybridReplay, ...
    target, control, 'Grid-first hybrid');
```

Grid-first hybrid: verified. p = 0.019375, n = 3, replay error = 0.000563047, search-replay divergence = 0
Note: the divergence is exactly zero, so the replay repeated the search's
own propagation rather than recomputing it independently. The search and the
replay reach the section through the same function, so agreement is guaranteed
and is not evidence that the returned perturbation is correct. The perturbation
is not thereby wrong; this particular check simply carries no weight for it.

```
if bisectionSearch.horizon ~= hybridSearch.horizon
    fprintf(['Note: the methods selected different horizons (%d and %d). ', ...
        'Perturbation magnitudes are not directly comparable at ', ...
        'different horizons.\n'], bisectionSearch.horizon, hybridSearch.horizon);
end
```

Controlled Replay and Verification

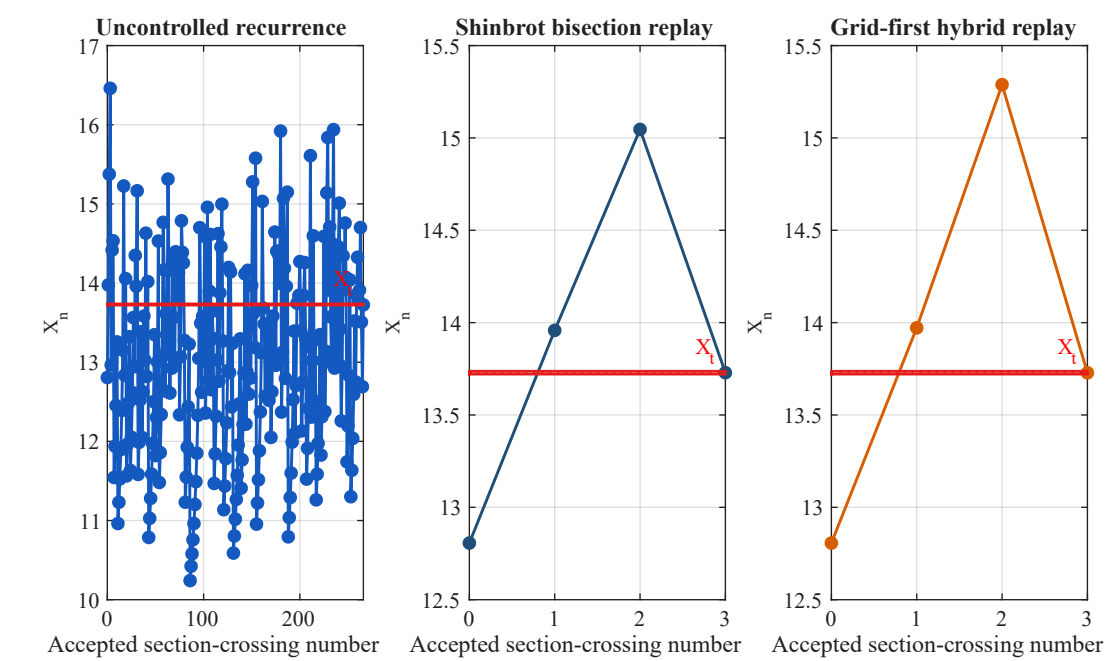
The selected controls are replayed independently of the search calculations. A method is reported as successful here only if the reintegrated controlled trajectory enters the target neighbourhood at the selected accepted crossing.

```
replayVerification = table( ...
    ["Shinbrot bisection"; "Grid-first hybrid"], ...
    [bisectionReplay.hit; hybridReplay.hit], ...
    [bisectionReplay.numCrossings; hybridReplay.numCrossings], ...
    [bisectionReplay.targetError; hybridReplay.targetError], ...
    [bisectionReplay.replayTime; hybridReplay.replayTime], ...
    'VariableNames', {'Method', 'ReplayHit', 'ReplayCrossings', ...
        'ReplayTargetError', 'ReplayTime'});
disp(replayVerification);
```

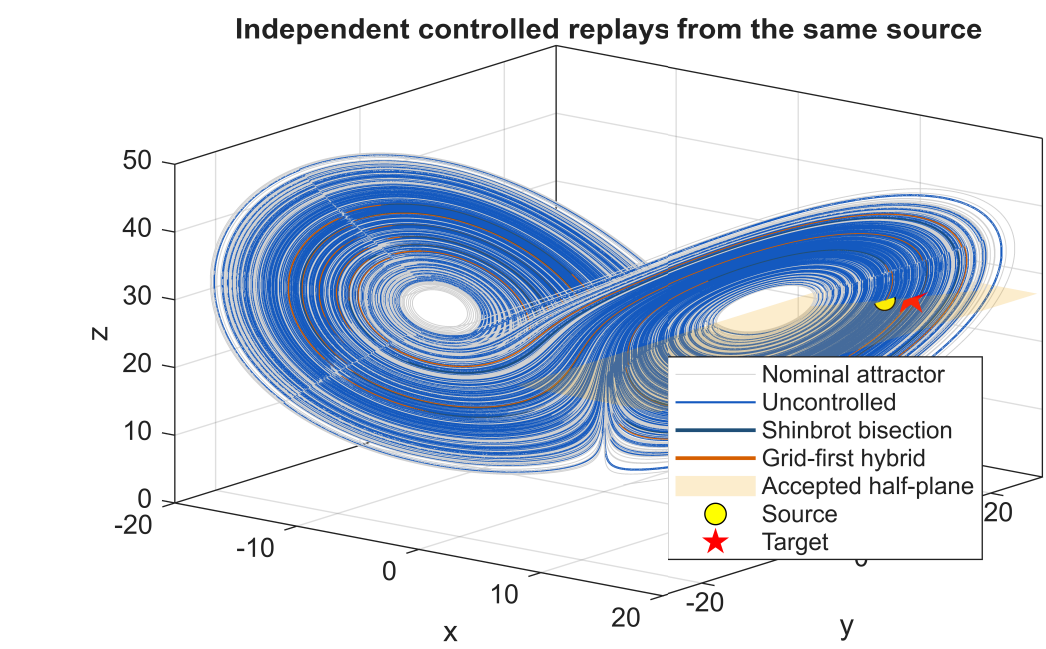
Method	ReplayHit	ReplayCrossings	ReplayTargetError	ReplayTime
"Shinbrot bisection"	true	3	4.0648e-09	0.071135
"Grid-first hybrid"	true	3	0.00056305	0.048091

```
walkthrough_plot_section_sequences( ...
```

```
uncontrolled, bisectionReplay, hybridReplay, target);
```



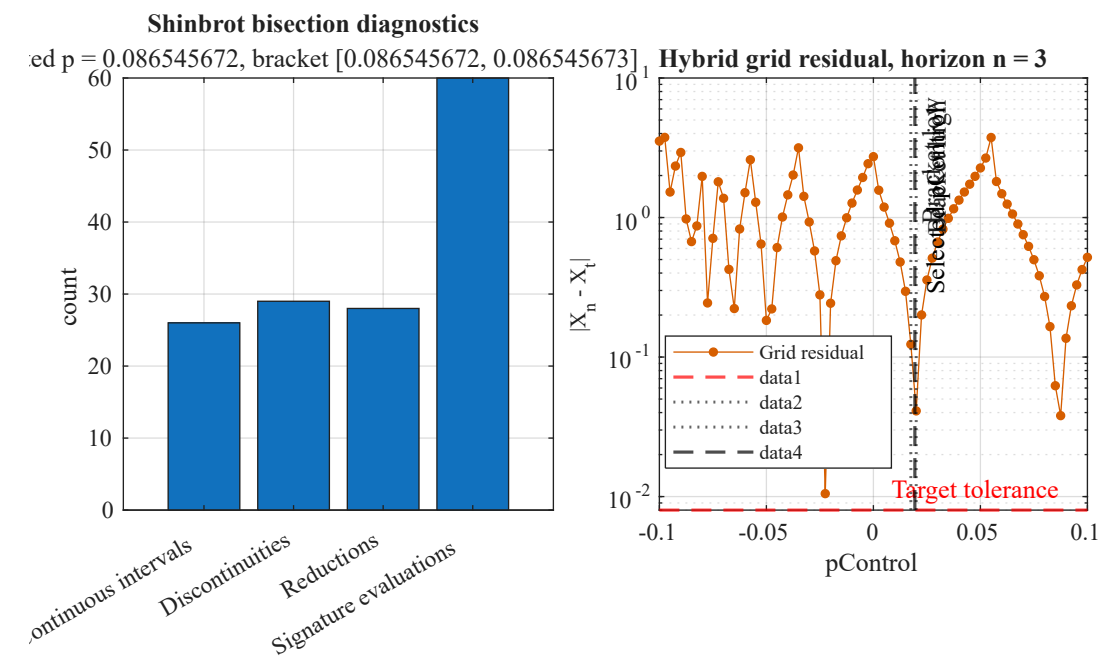
```
walkthrough_plot_controlled_phase_space( ...
    mapData, uncontrolled, bisectionReplay, hybridReplay, ...
    sourceState, target, params);
```



Search Behaviour Figures

The bisection method is represented by its final bracket and counters, while the hybrid method has a complete residual curve at the selected horizon. The figure below displays both forms of search evidence without replacing either algorithm.

```
walkthrough_plot_search_behaviour( ...
    bisectionSearch, hybridSearch, target, control);
```



RMS Scale and Timing Benchmark

The time-weighted RMS of the nominal vector field gives a scale for the perturbation in the Lorenz equations. The five-repetition timing benchmark times the two deterministic search-and-replay workflows from the already selected source state.

```
fprintf('Computing the post-transient time-weighted RHS RMS...\n');

Computing the post-transient time-weighted RHS RMS...
```

```
rhsRms = walkthrough_compute_rhs_rms(mapData, params, control, 100);
disp(struct2table(rhsRms));

    rmsFx      rmsFy      rmsFz      transientCrossingIndex      transientTime      averagingDuration      maxParameterEffect      parameterToRmsFyRatio      paperApproximateRmsFy      rmsFyDifferenceFromPaperApproximation

    42.731      65.409      78.84           100           161.28           1670.5           0.1           0.0015288           100           -34.591

fprintf('Benchmarking the deterministic bisection and hybrid methods...\n');

Benchmarking the deterministic bisection and hybrid methods...

searchBenchmark = benchmark_shinbrot_search_methods( ...
    sourceState, target, params, control, benchmarkSettings);
disp(searchBenchmark.comparisonTable);
```

Sixty-Source Ensemble Benchmark

The deterministic ensemble uses 60 source states chosen from accepted return-map indices 100:300, with scalar separation ``abs(X_s-X_t) >= 1.5``. Each selected source is tested once with the Shinbrot bisection method and once with the hybrid method. The saved result is accepted only when its metadata match the current target, control settings, method names, source count and table schema.

Regeneration is deliberately controlled by ``runExpensiveEnsemble`` because the existing verified ``mat`` and ``csv`` files are the current numerical authority for the ensemble section.

```
sourceEnsembleSettings.candidateIndices = 100:300;
sourceEnsembleSettings.numSources = 60;
sourceEnsembleSettings.sourceSeparation = 1.5;
sourceEnsembleSettings.outputMatFile = 'shinbrot_source_ensemble.mat';
sourceEnsembleSettings.methodResultsCsv = ...
    'shinbrot_source_ensemble_method_results.csv';
sourceEnsembleSettings.pairedResultsCsv = ...
    'shinbrot_source_ensemble_paired_results.csv';

if runExpensiveEnsemble
    sourceEnsemble = benchmark_shinbrot_source_ensemble( ...
        mapData, target, params, control, sourceEnsembleSettings);
else
    savedEnsemble = load(sourceEnsembleSettings.outputMatFile);
    if ~walkthrough_is_valid_source_ensemble(savedEnsemble, target, control)
        error(['Saved 60-source ensemble results are missing or stale. ', ...
            'Set runExpensiveEnsemble = true to regenerate them.']);
    end
    sourceEnsemble = savedEnsemble;
end

disp(sourceEnsemble.methodSummary);
```

Method	NumberSources	NumberReplayHits	SuccessRate	MeanHorizon	MedianHorizon	MeanAbsoluteP	MedianAbsoluteP	MeanReplayTargetError	MedianReplayTargetError	MaximumReplayTargetError	MeanSearchTime	MedianSearchTime
{'shinbrotPaperBisection'}	60	60	1	5.65	5	0.054252	0.054624	2.4657e-08	1.7028e-08	9.3811e-08	1.5432	1.3204
{'hybridGridBisection' }	60	60	1	5.6667	5	0.036932	0.021221	0.0038148	0.0036413	0.0079304	12.342	11.16

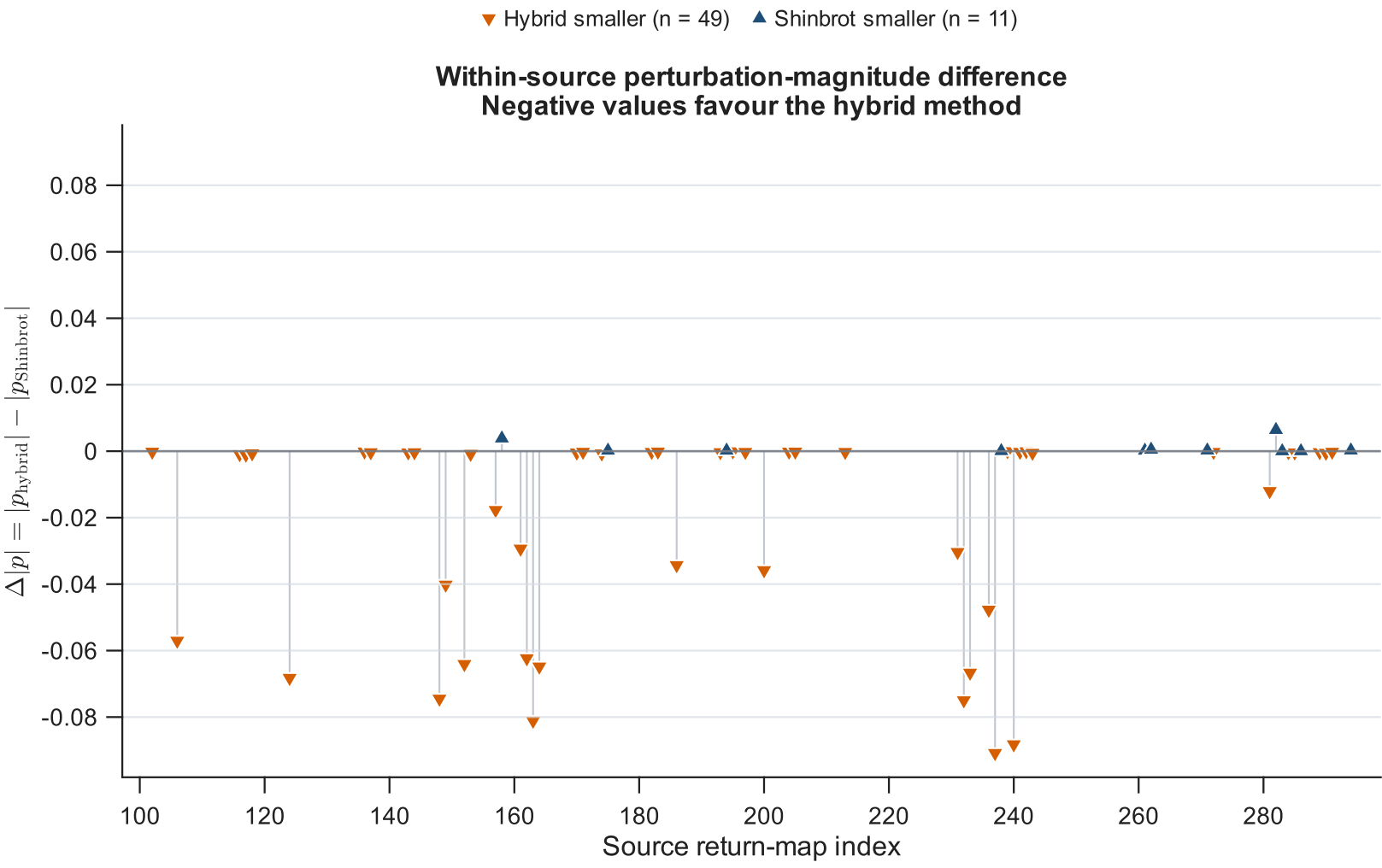
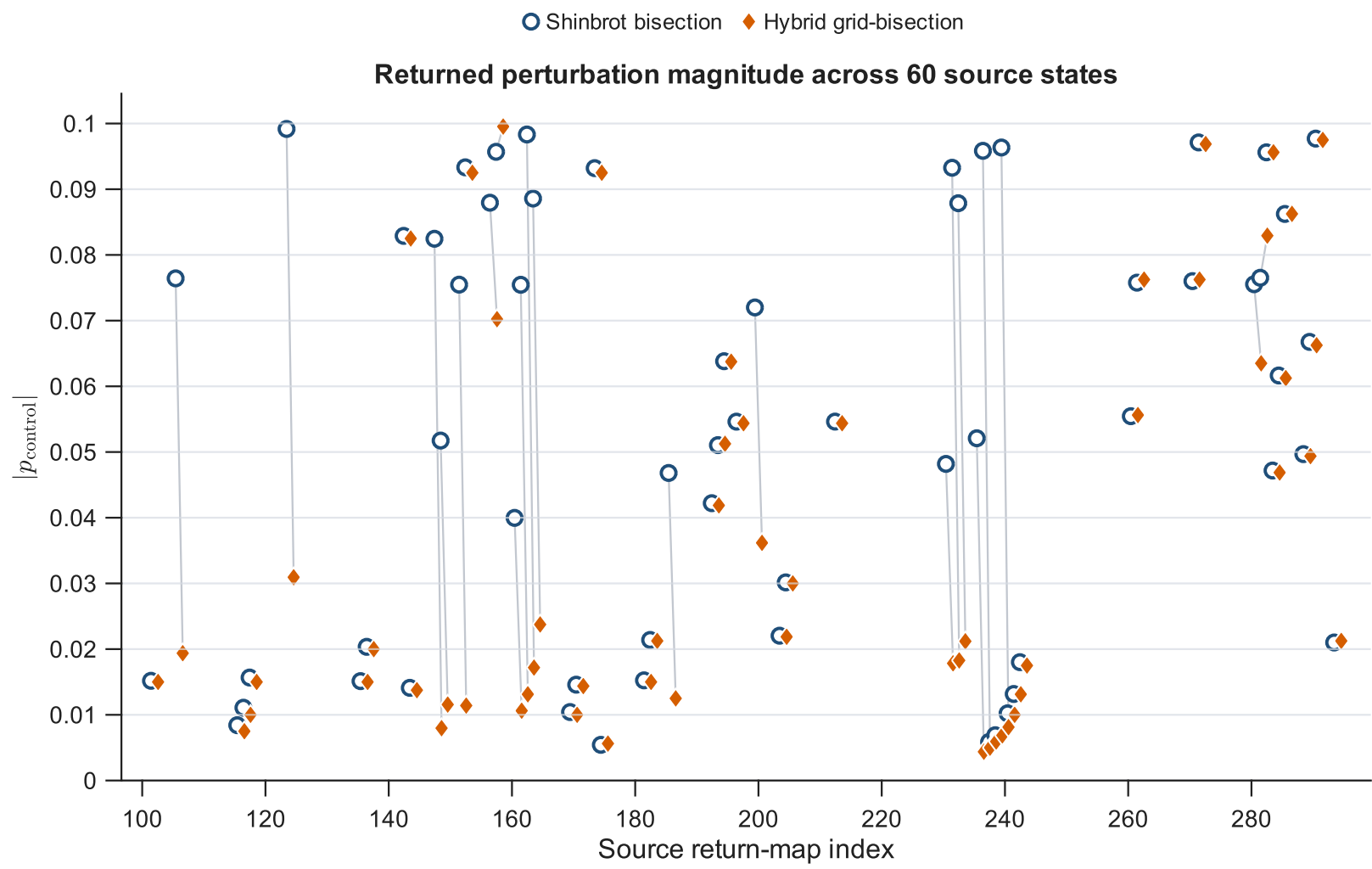
```
disp(sourceEnsemble.summary);
```

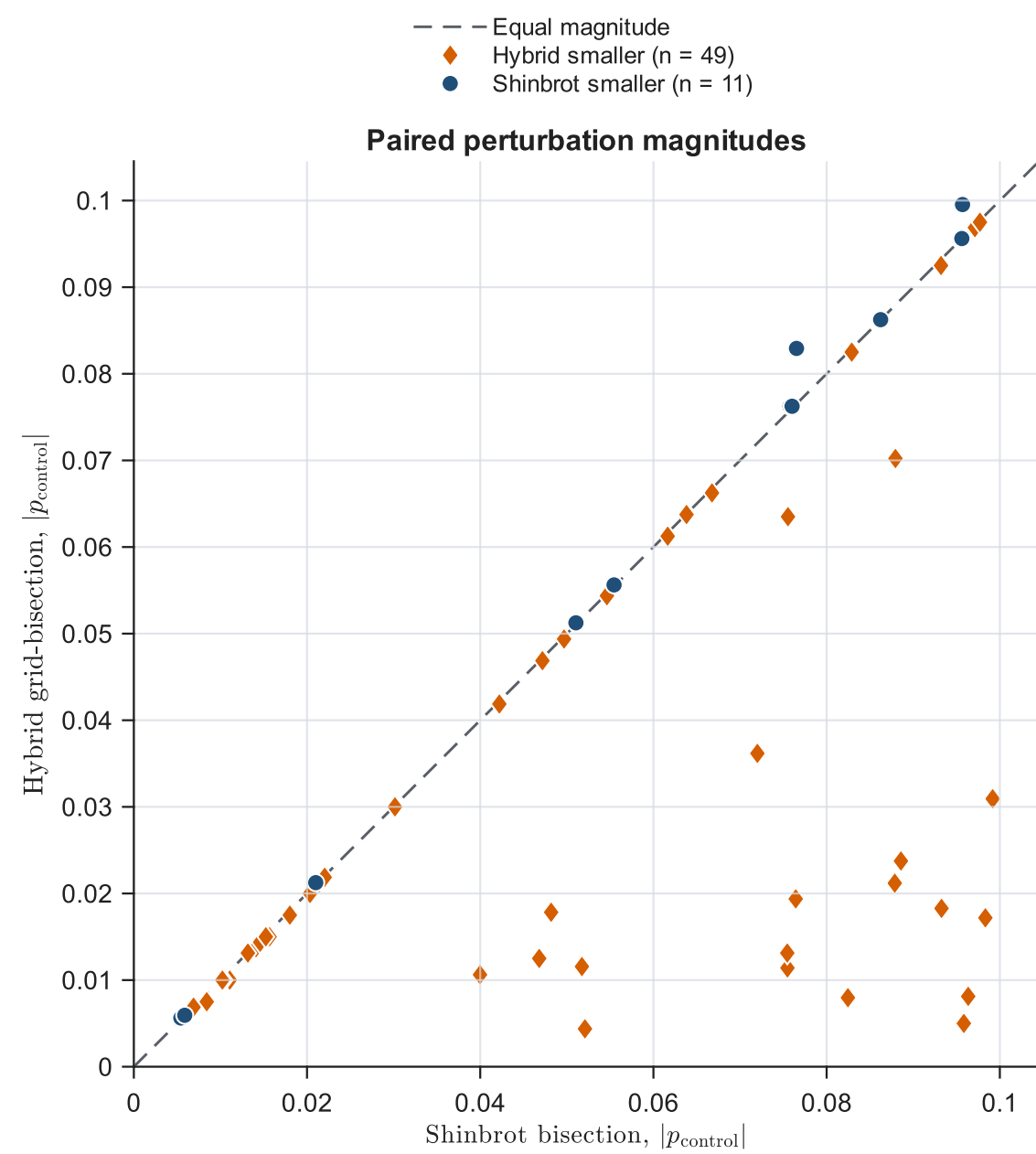
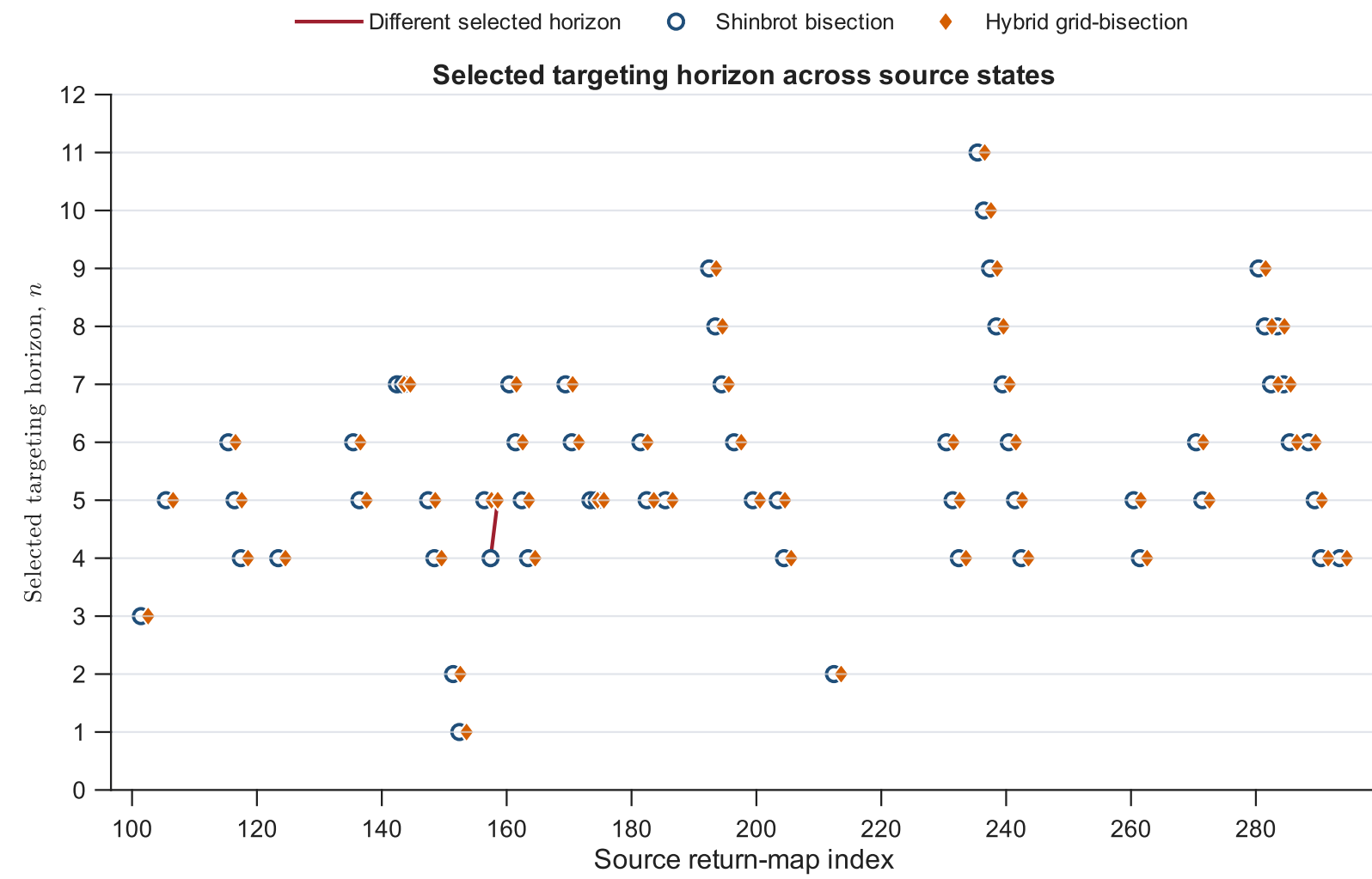
NumberSourcesTested	NumberShinbrotReplayHits	NumberHybridReplayHits	ShinbrotSuccessRate	HybridSuccessRate	NumberBothReplayHit	NumberSameSuccessfulHorizon	NumberDifferentSuccessfulHorizons	MatchedHybridSmaller	MatchedShinbrotSmaller	MatchedTies	
60	60	60	1	1	60	59	1	49	10	0	{[102 106 116 117 118 124 136 137 143 144 148 149 1

```
ensembleExcerpt = sourceEnsemble.pairedResultsTable(1:min(12, ...
    height(sourceEnsemble.pairedResultsTable)), :);
disp(ensembleExcerpt);
```

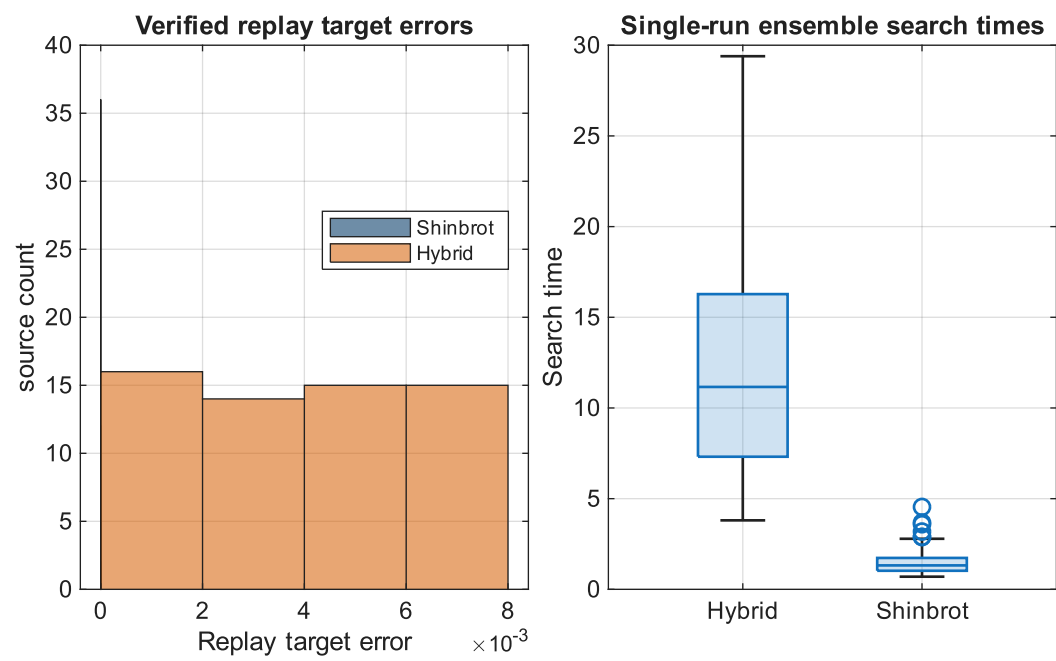
SourceIndex	SourceX	ShinbrotHit	HybridHit	ShinbrotReplayHit	HybridReplayHit	ShinbrotP	HybridP	ShinbrotAbsoluteP	HybridAbsoluteP	AbsolutePDifference	ShinbrotHorizon	HybridHorizon	ShinbrotFinalTargetError	HybridFinalTargetError	ShinbrotReplayTargetError	HybridReplayTargetError
102	15.268	true	true	true	true	-0.015177	-0.015	0.015177	0.015	-0.00017705	3	3	3.0979e-09	0.0042612	3.098e-09	0.0042612
106	12.045	true	true	true	true	-0.076431	0.019375	0.076431	0.019375	-0.057056	5	5	1.5193e-08	0.00043493	1.5192e-08	0.00043493
116	11.297	true	true	true	true	-0.0084194	-0.0075	0.0084194	0.0075	-0.00091941	6	6	3.3106e-08	0.0069031	3.3106e-08	0.0069031
117	11.63	true	true	true	true	-0.011078	-0.01	0.011078	0.01	-0.0010785	5	5	2.5502e-08	0.0061548	2.5502e-08	0.0061548
118	12.031	true	true	true	true	-0.015674	-0.015	0.015674	0.015	-0.00067429	4	4	3.9873e-09	0.0027218	3.9874e-09	0.0027218
124	12.149	true	true	true	true	-0.099172	-0.030937	0.099172	0.030937	-0.068234	4	4	2.7722e-08	0.0015528	2.7721e-08	0.0015528
136	11.419	true	true	true	true	0.015127	0.015	0.015127	0.015	-0.00012671	6	6	1.0513e-08	0.0024236	1.0513e-08	0.0024236
137	11.78	true	true	true	true	0.020347	0.02	0.020347	0.02	-0.00034707	5	5	3.4678e-08	0.0049381	3.4678e-08	0.0049381
143	11.258	true	true	true	true	-0.082898	-0.0825	0.082898	0.0825	-0.00039836	7	7	4.4106e-08	0.005186	4.4106e-08	0.005186
144	11.582	true	true	true	true	0.014107	0.01375	0.014107	0.01375	-0.0003567	7	7	3.4839e-09	0.0058741	3.4839e-09	0.0058741
148	11.728	true	true	true	true	0.082458	0.0079688	0.082458	0.0079688	-0.074489	5	5	3.1149e-08	0.0014675	3.115e-08	0.0014675
149	12.178	true	true	true	true	0.051743	0.011562	0.051743	0.011562	-0.040181	4	4	3.7881e-08	0.0028989	3.788e-08	0.0028989

```
plot_shinbrot_source_ensemble(sourceEnsemble.pairedResultsTable);
```





```
walkthrough_plot_ensemble_distributions(sourceEnsemble);
```



Noise-Robustness Investigation

The final investigation uses fixed-step RK4 because adaptive internal steps would make per-step coordinate noise ambiguous. In `next_valid_section_crossing_noisy_rk4`, independent Gaussian noise is applied before the corresponding integration step:

$$X \leftarrow X + \sigma_{\text{noise}}\eta, \quad \eta \sim N(0, I_3).$$

Section crossings are detected manually by linear interpolation across the integrated RK4 segment. Crossings caused solely by the instantaneous noise jump are deliberately excluded. The final noise study uses the verified bisection method only. The retained mode is `sectionRetargetMode`: after each accepted section crossing that misses the target, the verified bisection search is rerun from that accepted section state. This differs from a one-time targeting experiment and is also not a full reproduction of every recovery mechanism discussed by Shinbrot et al.

```

noise.dt = 0.001;
noise.maxTimeToNextValidSection = 80;
noise.maxControlledCrossings = 60;
noise.maxRejectedCrossings = 250;
noise.maxStateNorm = 1e5;
noise.maxRk4Steps = 12000;
noise.axisDefinition = ...
    'per-step per-coordinate Gaussian standard deviation';
noise.directlyComparableToPaperFigure6 = false;

noiseControl = control;
noiseControl.searchMethod = verifiedBisectionIdentifier;

noiseSweep = walkthrough_noise_sweep_settings( ...
    verifiedBisectionIdentifier, sourceIndex, runNoiseSmokeOnly);

if runExpensiveNoiseSweep
    noiseResults = main_shinbrot_noise_demo(true, runNoiseSmokeOnly);
else
    currentNoiseConfig = build_bisection_noise_config( ...
        sourceState, target, params, noiseControl, noise, noiseSweep);
    [noiseResults, savedNoiseLoaded, noiseDifferences] = ...
        load_valid_bisection_noise_results( ...
            noiseSweep.outputMatFile, currentNoiseConfig);
    if ~savedNoiseLoaded
        error(['No valid completed bisection-noise result was loaded. ', ...
            'Set runExpensiveNoiseSweep = true to regenerate. ', ...
            'Validation differences: %s'], ...
            strjoin(cellstr(noiseDifferences), ' '););
end
end

```

```

===== Paper-Bisection Noisy Shinbrot Lorenz Targeting =====
Generating deterministic source state X_150...
Source X_s = 12.8066889599, target X_t = 13.729
Mean return time Tbar = 1.51859, steps per mean return = 1518.6, 40-step time = 0.04, cadence ratio = 0.02634
Default full-trial RK4-step budget = 227789 (safety factor 2.5).
Finding initial deterministic bisection perturbation...
Initial pControl = 0.0865456722677, horizon = 3
Active RK4-step budget = 10631, active crossing limit = 5.

```

===== Open-Loop Zero-Noise Solver Comparison =====													
selectedP	horizon	adaptiveFinalX	fixedStepFinalX	adaptiveTargetError	fixedStepTargetError	absoluteSolverDiscrepancy	discrepancyFractionOfTolerance	rk4StepCount	acceptedCrossingCount	rejectedCrossingCount	adaptiveHit	fixedStepHit	fixedStepFailureReason
0.086546	3	13.729	13.712	4.0648e-09	0.017418	0.017418	2.1773	6099	3	5	true	false	{0x0 char}
Warning: Configured-dt fixed-step open-loop replay did not hit the target with the same constant pControl. Running a reduced timestep convergence diagnostic before any sweep.													
horizon	selectedP	fixedStepFinalX	fixedStepTargetError	absoluteSolverDiscrepancy	discrepancyFractionOfTolerance	rk4StepCount	acceptedCrossingCount	rejectedCrossingCount	fixedStepHit				
3	0.086546	13.712	0.017418	0.017418	2.1773	6099	3	5	false				
3	0.086546	13.724	0.005139	0.005139	0.64237	12199	3	5	true				
3	0.086546	13.728	0.00053169	0.00053169	0.006462	24395	3	5	true				

=== Zero-noise acceptance gate ===
implementation : shinbrotPaperBisection
source index 150, X_s = 12.806689, X_t = 13.729, eps_t = 0.008
dt = 0.001, maxControlledCrossings = 5, maxTimeToNextValidSection = 20
mean return time of the nominal map is about 1.519, so one
accepted crossing costs roughly 1519 fixed steps.

deterministic reference (adaptive solver)
p : 0.0865456722677
horizon : 3
replay error : 4.06482e-09

paper40StepRetargetMode
reached target : 1
crossings to hit : 3 (deterministic horizon 3)
accepted crossings: 3
final error : 7.67317e-05
retargets fired : 152
control unchanged : 0
RK4 steps : 6099
passed

mode	passed	hit	crossingsToTarget	acceptedCrossings	deterministicHorizon	fixedStepFinalError	adaptiveFinalError	discretisationDifference	discretisationFractionOfTolerance	retargetsFired	searchEvaluations	controlUnchanged	rk4Steps	stepBudgetExhausted	integratio
"paper40StepRetargetMode"	true	true	3	3	3	7.6732e-05	4.0648e-09	7.6728e-05	0.0095909	152	153	false	6099	false	fals

modes passed : 1 of 1

Gate passed. The discretisation difference above is the adaptive
versus fixed-step discrepancy that Subsection 3.1.9 promises to report,
measured from the same source and the same perturbation.

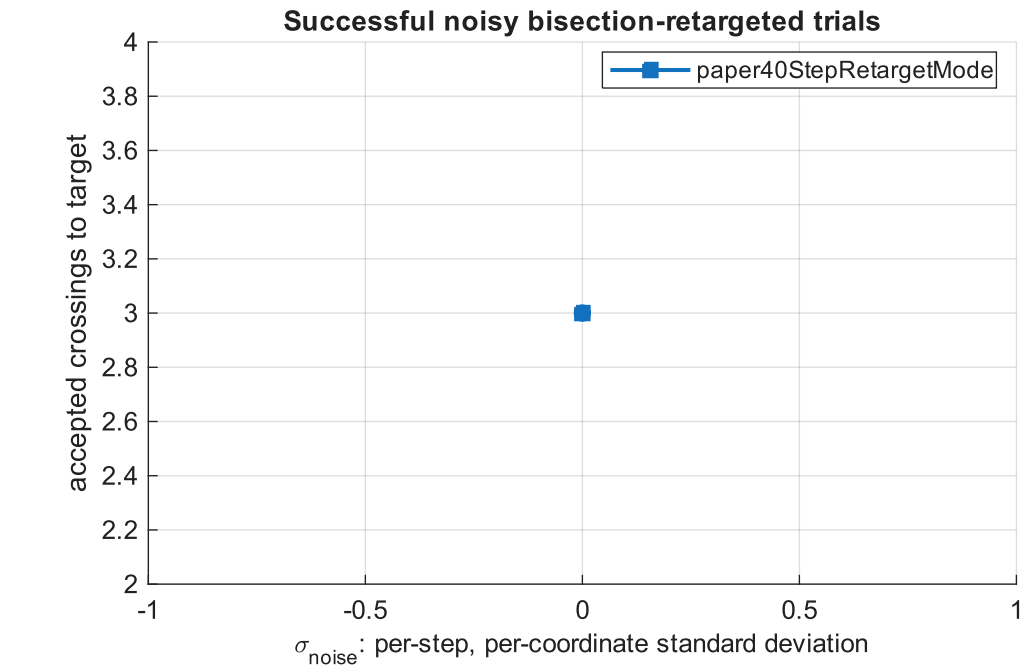
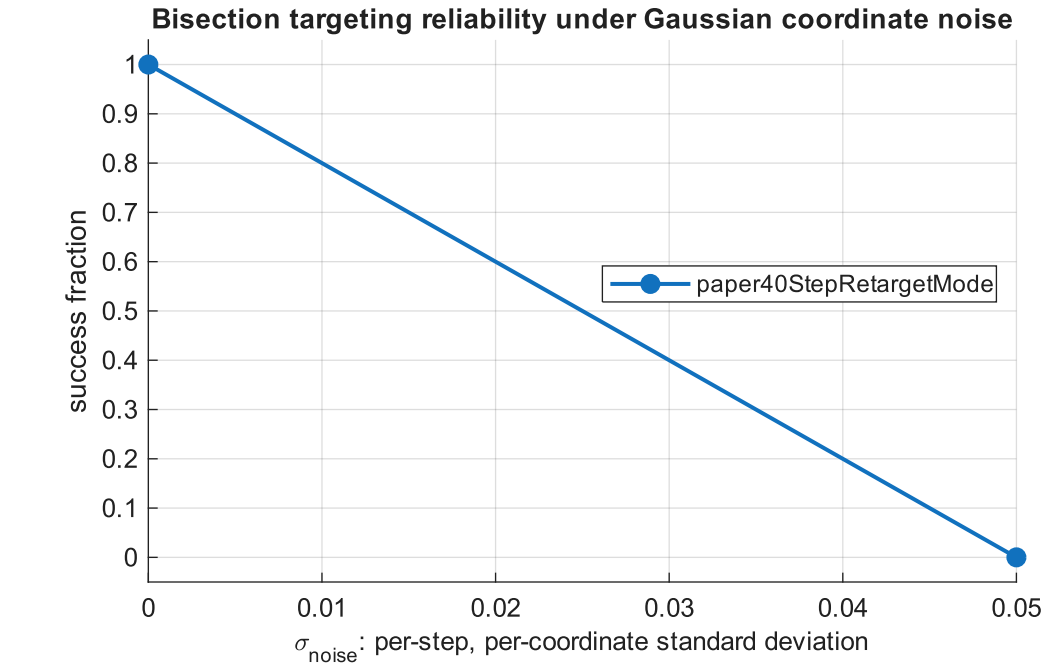
=====

Running reduced bisection noise smoke test (2 trial(s))...

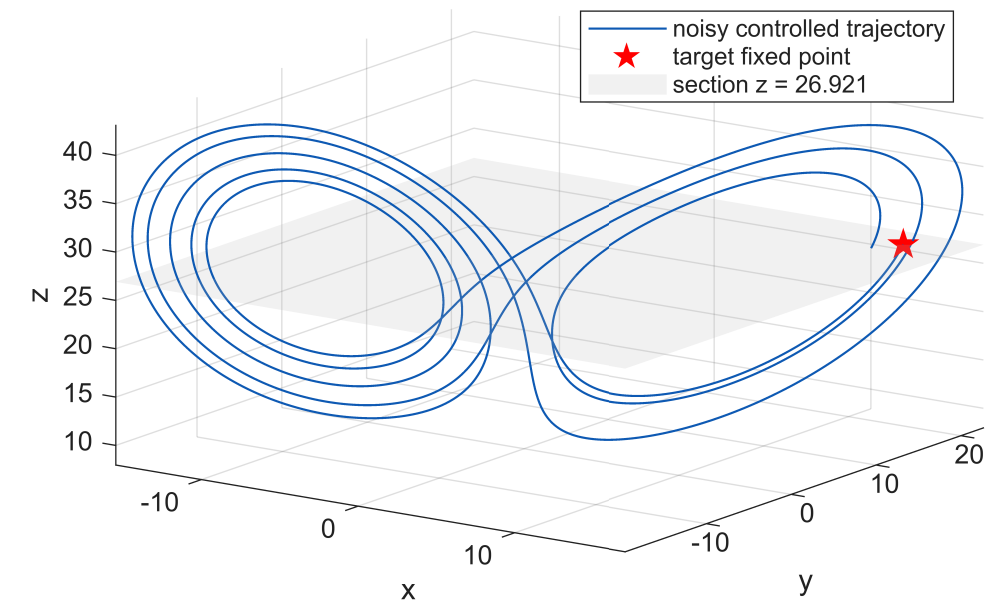
Trial 1/2 | sigma_noise = 0 | realization 1/1 | success = 1 | elapsed = 1.8 min | ETA = 1.8 min

Trial 2/2 | sigma_noise = 0.05 | realization 1/1 | success = 0 | elapsed = 6.8 min | ETA = 0.0 min

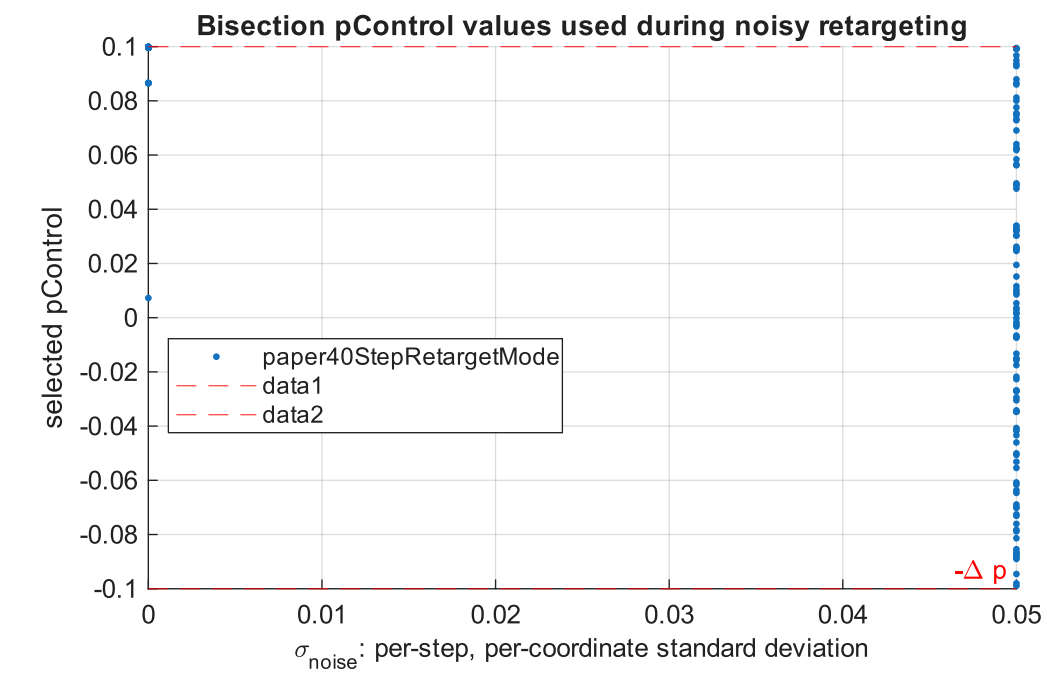
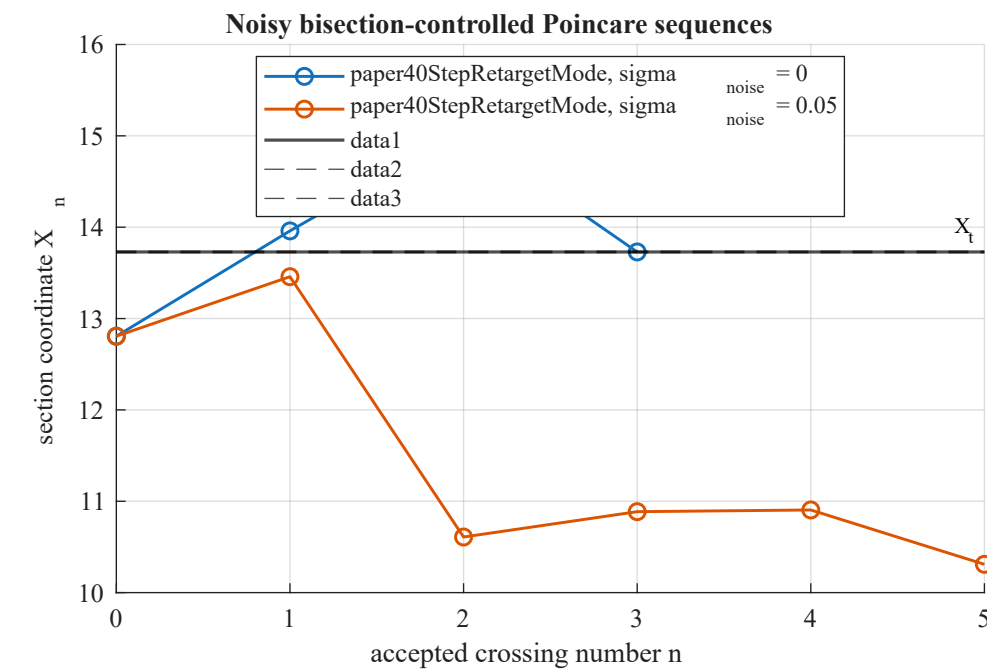
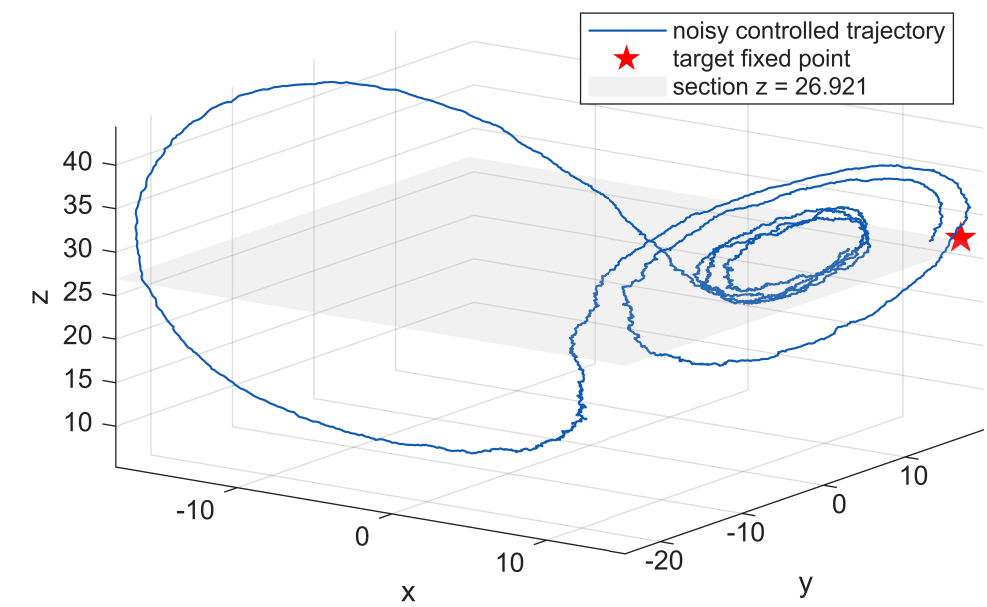
===== Paper-Bisection Gaussian-Noise Summary =====													
Mode	SigmaNoise	Successes	TotalTrials	SuccessFraction	WilsonLower95	WilsonUpper95	MeanCrossingsToTargetAmongSuccesses	MedianCrossingsToTargetAmongSuccesses	MeanAcceptedCrossingsAllTrials	MeanFinalTargetError	MedianFinalTargetError	StdFinalTargetError	MeanRk4Steps
{ 'paper40StepRetargetMode' }	0	1	1	1	0.20655	1	3	3	3	7.6732e-05	7.6732e-05	NaN	6099
{ 'paper40StepRetargetMode' }	0.05	0	1	0	0	0.79345	NaN	NaN	5	3.4195	3.4195	NaN	4327



successful bisection-controlled trajectory (paper40StepRetargetMode, sigma_{noise}



near or marginal bisection-controlled trajectory (paper40StepRetargetMode, sigma_{noise}



Noise plots use per-step, per-coordinate Gaussian standard deviation and fixed RK4 step $dt = 0.001$.
Directly comparable to Shinbrot Fig. 6 axis: false
Verified bisection identifier: shinbrotPaperBisection
Noise retargeting mode: paper40StepRetargetMode
Total configured trials: 2

Result completed: true
Smoke test: true
Directly comparable to Shinbrot Fig. 6 axis: false

```
noiseCompletion = table(string(noiseResults.searchMethod), ...
    noiseResults.completed, noiseResults.isSmokeTest, ...
    noiseResults.totalTrials, numel(noiseResults.sigmaNoiseValues), ...
    noiseResults.numTrials, string(noiseResults.modes{1}), ...
    string(noiseResults.noiseAxisDefinition), ...
    noiseResults.directlyComparableToPaperFigure6, ...
    'VariableNames', {'SearchMethod', 'Completed', 'IsSmokeTest', ...
    'TotalTrials', 'NumberNoiseLevels', 'TrialsPerLevel', ...
    'RetargetingMode', 'NoiseAxisDefinition', ...
    'DirectlyComparableToPaperFigure6'});
disp(noiseCompletion);
```

SearchMethod	Completed	IsSmokeTest	TotalTrials	NumberNoiseLevels	TrialsPerLevel	RetargetingMode	NoiseAxisDefinition	DirectlyComparableToPaperFigure6
"shinbrotPaperBisection"	true	true	2	2	1	"paper40StepRetargetMode"	"per-step per-coordinate Gaussian standard deviation"	false

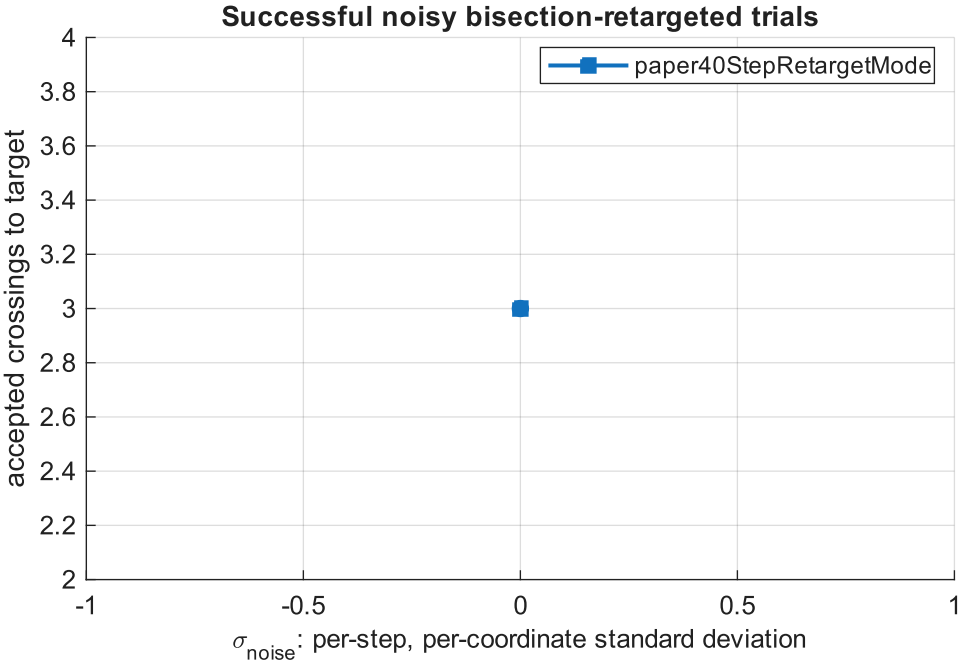
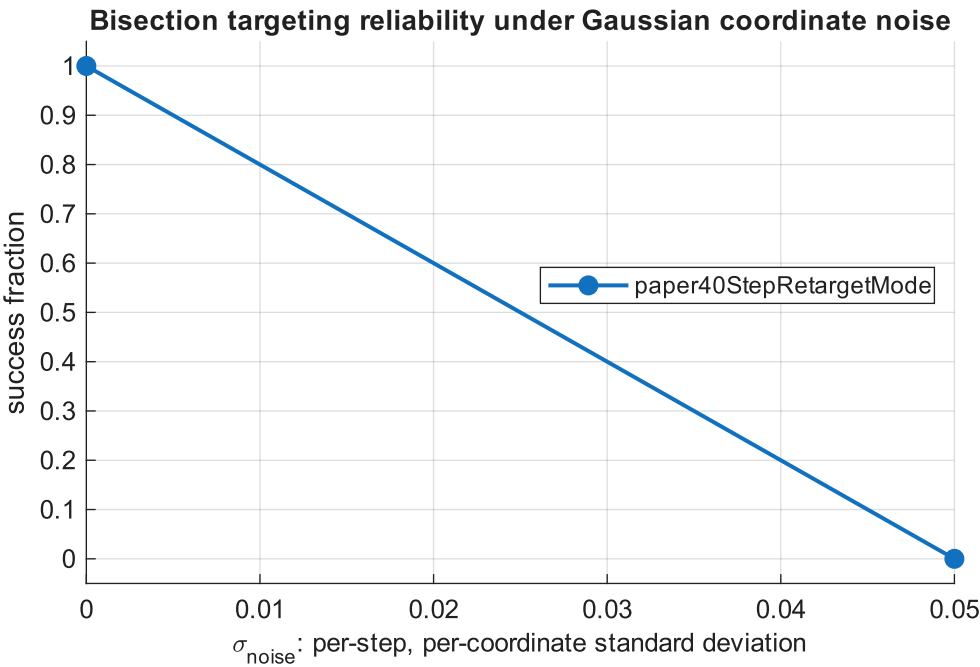
```
disp(noiseResults.summaryTable);
```

Mode	SigmaNoise	Successes	TotalTrials	SuccessFraction	WilsonLower95	WilsonUpper95	MeanCrossingsToTargetAmongSuccesses	MedianCrossingsToTargetAmongSuccesses	MeanAcceptedCrossingsAllTrials	MeanFinalTargetError	MedianFinalTargetError	StdFinalTargetError	MeanRk4Steps
{'paper40StepRetargetMode'}	0	1	1	1	0.20655	1	3	3	3	7.6732e-05	7.6732e-05	NaN	6099
{'paper40StepRetargetMode'}	0.05	0	1	0	0	0.79345	NaN	NaN	5	3.4195	3.4195	NaN	4327

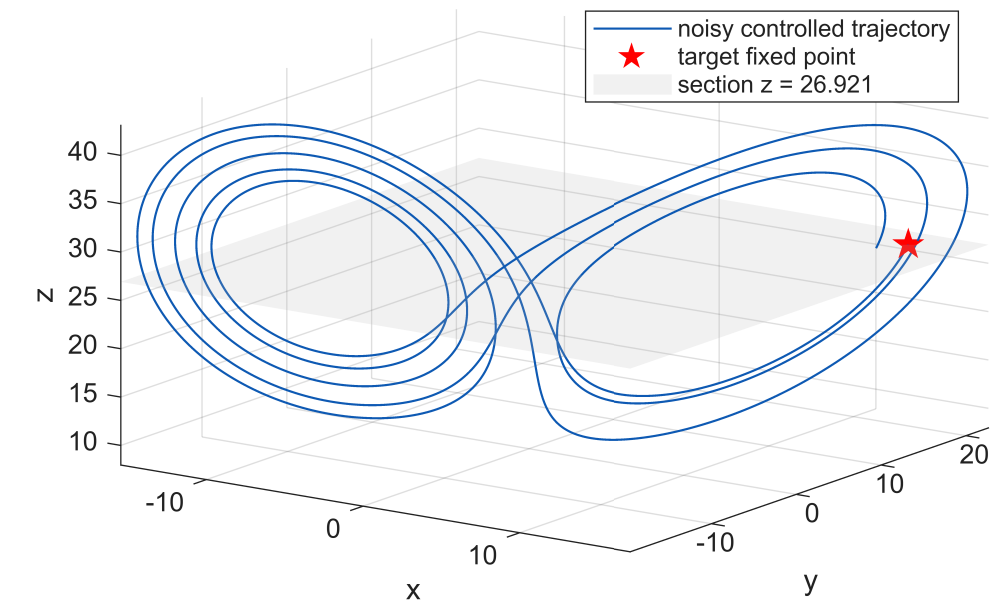
```
failureRows = noiseResults.trialsTable(~noiseResults.trialsTable.Hit, :);
failureSummary = table(height(failureRows), ...
    sum(noiseResults.trialsTable.IntegrationFailed), ...
    'VariableNames', {'NumberFailedTargetingTrials', ...
    'NumberIntegrationFailures'});
disp(failureSummary);
```

NumberFailedTargetingTrials	NumberIntegrationFailures
1	0

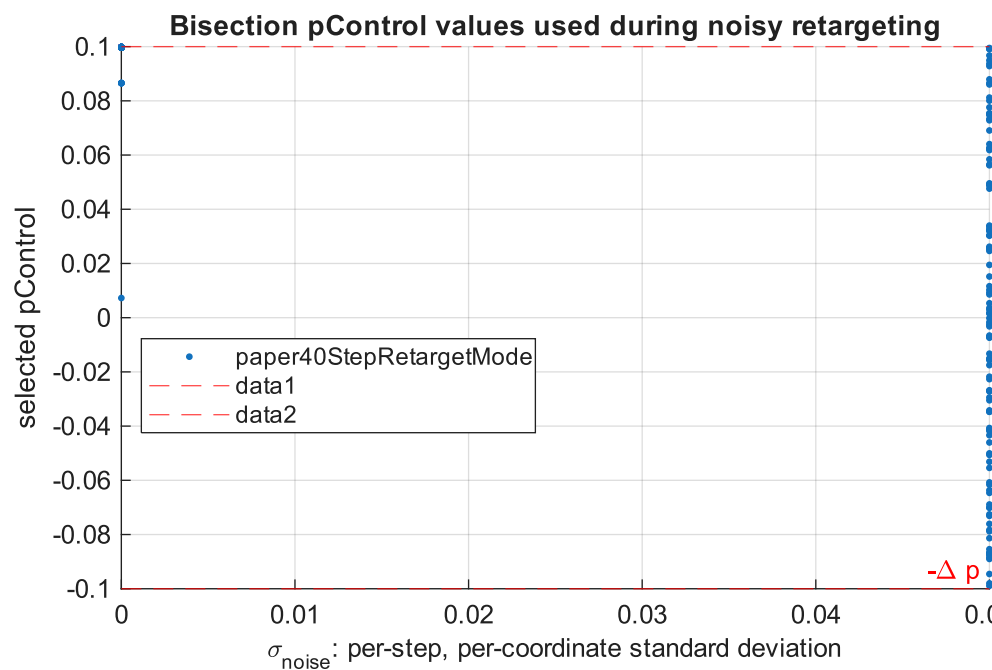
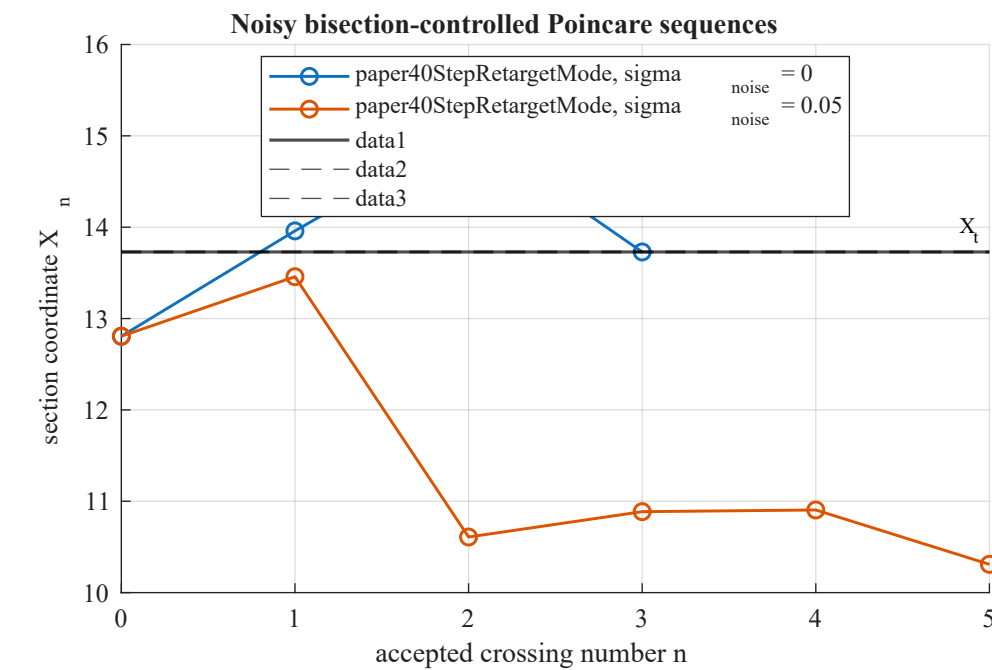
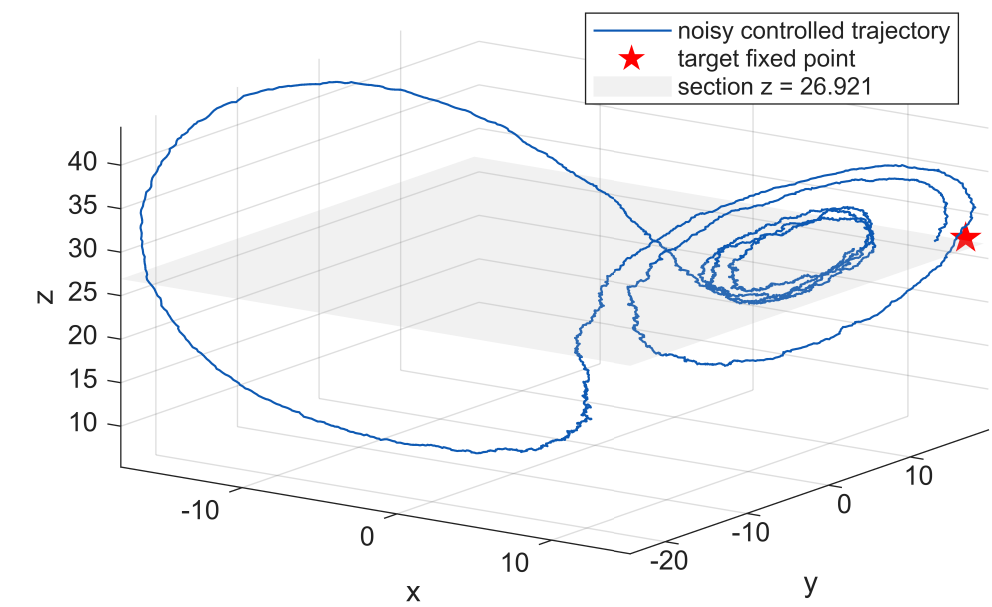
```
plot_noise_results(noiseResults, target, params, noise);
```



successful bisection-controlled trajectory (paper40StepRetargetMode, sigma_{noise} = 0)

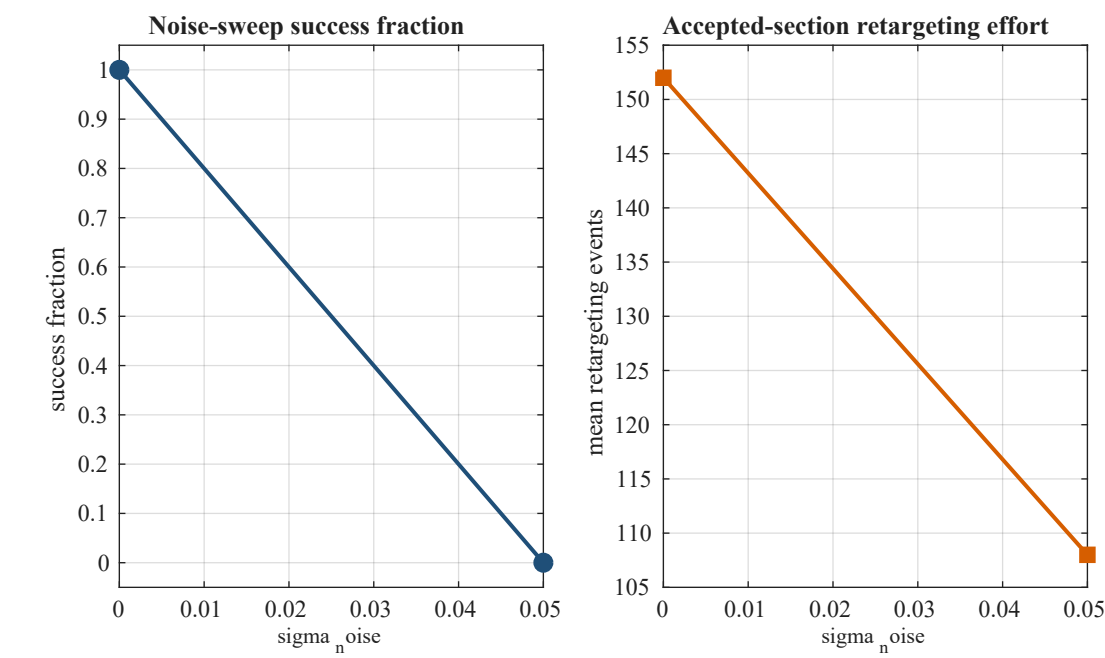


near or marginal bisection-controlled trajectory (paper40StepRetargetMode, sigma_{noise} = 0.05)



Noise plots use per-step, per-coordinate Gaussian standard deviation and fixed RK4 step $dt = 0.001$.
Directly comparable to Shinbrot Fig. 6 axis: false

```
walkthrough_plot_noise_summary(noiseResults);
```



Scaling in the Control Bound

The fourth component tests the scaling relation $n_t \sim \ln(1/\text{dp})/\lambda$, which is the only quantitative theoretical prediction of the original paper. The maximum admissible perturbation is varied over the decades from $1\text{e-}2$ to 1 , twenty-five randomly chosen source states are directed to the target at each value, and the mean number of accepted crossings is regressed on $\log_{10}(\text{dp})$. The fitted slope is converted to an exponent through $\lambda = \ln(10)/|s|$ and compared both with the published fit and with the exponent measured directly from the nominal return map. The horizon cap is raised for this component, because a cap that binds at small dp would truncate the trend being measured, and every trial that terminates at the cap is counted so that a binding cap is visible. The sweep is expensive, so a saved result is loaded unless `runControlBoundSweep` is set.

```
controlBoundSweepFile = 'shinbrot_control_bound_sweep.mat';

if runControlBoundSweep
    controlBoundResults = shinbrot_control_bound_sweep(mapData, ...
        struct('saveMatFile', controlBoundSweepFile));
elseif isfile(controlBoundSweepFile)
    loadedControlBound = load(controlBoundSweepFile, 'results');
    controlBoundResults = loadedControlBound.results;
    fprintf('Loaded saved control-bound sweep from %s.\n', controlBoundSweepFile);
else
    error(['No saved control-bound sweep was found. ', ...
        'Set runControlBoundSweep = true to generate it.']);
end
```

Loaded saved control-bound sweep from shinbrot_control_bound_sweep.mat.

disp(controlBoundResults.levels);											
deltaP	log10DeltaP	sources	successes	successFraction	meanHorizon	medianHorizon	stdHorizon	minHorizon	maxHorizon	trialsAtHorizonCap	seconds
0.01	-2	25	25	1	8	8	2.3805	5	13	0	42.124
0.017783	-1.75	25	25	1	6.92	7	2.1393	4	11	0	36.079
0.031623	-1.5	25	25	1	6.52	6	2.0841	4	11	0	36.119
0.056234	-1.25	25	25	1	6.44	6	2.0224	4	11	0	32.193
0.1	-1	25	25	1	6.12	6	2.0273	4	11	0	32.106
0.17783	-0.75	25	25	1	5.8	5	2.0207	4	11	0	36.866
0.31623	-0.5	25	25	1	5.52	5	1.9391	4	10	0	37.816
0.56234	-0.25	25	25	1	4.84	4	1.7243	2	9	0	28.392
1	0	25	25	1	4.28	4	1.7445	2	9	0	20.611

```
controlBoundSummary = table( ...
    controlBoundResults.fit.slope, ...
    controlBoundResults.fit.slopeStandardError, ...
    controlBoundResults.fit.rSquared, ...
    controlBoundResults.fit.lambdaFromSlope, ...
    controlBoundResults.published.slope, ...
    controlBoundResults.returnMapLambda, ...
    controlBoundResults.returnMapImpliedSlope, ...
    controlBoundResults.crossingRatio.ratio, ...
    controlBoundResults.lambdaPerPlaneCrossing, ...
    string(controlBoundResults.classification), ...
    controlBoundResults.capBound, ...
    'VariableNames', {'FittedSlope', 'SlopeStandardError', 'RSquared', ...
    'LambdaFromSlope', 'PublishedSlope', 'ReturnMapLambda', ...
    'ReturnMapImpliedSlope', 'PlaneToAcceptedRatio', ...
    'LambdaPerPlaneCrossing', 'Classification', 'HorizonCapBound'});
disp(controlBoundSummary);
```

FittedSlope	SlopeStandardError	RSquared	LambdaFromSlope	PublishedSlope	ReturnMapLambda	ReturnMapImpliedSlope	PlaneToAcceptedRatio	LambdaPerPlaneCrossing	Classification	HorizonCapBound
-1.584	0.13363	0.95255	1.4537	-2.49	1.5091	-1.5258	1.955	0.77193	"returnMapSlopeReproduced"	false

Interpretation and Limitations

The principal deterministic example verifies that both retained searches can target the selected fixed-point neighbourhood from the same source in three accepted crossings. In this example the revised Shinbrot bisection method finds the slightly smaller perturbation, while the hybrid method finds the smaller final residual. This single example does not establish universal superiority. The ensemble benchmark is the appropriate place to compare distributions of horizons, magnitudes, residuals and costs.

The hybrid method depends on grid resolution and on the rule used to refine adjacent brackets. The bisection method depends on how discontinuities in crossing identity are isolated and on the maximum isolation depth. Both methods depend on the Poincare section, target tolerance, ODE tolerances, source selection and maximum search horizon.

The noise study is a fixed-step, coordinate-noise experiment with accepted-section retargeting by the verified bisection method. Its noise axis is a per-step coordinate standard deviation in Lorenz units and is not directly comparable with every axis choice in the original paper. The results should therefore be interpreted as robustness diagnostics for this implementation rather than as a complete reproduction of all noisy recovery experiments.

```
interpretationTable = table( ...
  ["Principal example"; "60-source ensemble"; "Noise sweep"], ...
  ["Both deterministic methods replay into the target interval"; ...
    "Both methods attain verified hits across the saved source set"; ...
    "Bisection-only accepted-section retargeting is tested under RK4 noise"], ...
  ["Single-source comparison only"; ...
    "Summary statistics do not prove one method is universally better"; ...
    "Noise axis is implementation-specific and not directly paper-equivalent"], ...
  'VariableNames', {'StudyPart', 'SupportedFinding', 'Limitation'});
disp(interpretationTable);
```

StudyPart	SupportedFinding	Limitation
"Principal example"	"Both deterministic methods replay into the target interval"	"Single-source comparison only"
"60-source ensemble"	"Both methods attain verified hits across the saved source set"	"Summary statistics do not prove one method is universally better"
"Noise sweep"	"Bisection-only accepted-section retargeting is tested under RK4 noise"	"Noise axis is implementation-specific and not directly paper-equivalent"

Final Reproducibility Summary

The table below lists the main supporting '.m' files called by this Live Script. To rerun the principal demonstration, press Run in this Live Script or call the deterministic functions shown above. To regenerate the ensemble set `runExpensiveEnsemble = true`. To regenerate the full noise sweep set `runExpensiveNoiseSweep = true`; the default is to load the verified completed sweep.

```
supportingFiles = walkthrough_supporting_files_table(projectPath);
disp(supportingFiles);
```

FileName	Role	Exists
"lorenz_rhs.m"	"Lorenz vector field"	true
"poincare_event.m"	"Poincare plane event"	true
"next_valid_section_crossing.m"	"Accepted deterministic section crossing"	true
"generate_return_map.m"	"Nominal scalar return-map data"	true
"uncontrolled_hitting_time_1d.m"	"Uncontrolled recurrence baseline"	true
"search_parameter_to_target.m"	"Dispatcher for both deterministic searches"	true
"run_shinbrot_discontinuity_aware_bisection.m"	"Verified Shinbrot bisection search"	true
"evaluate_shinbrot_crossing_signature.m"	"Crossing-order signature diagnostic"	true
"find_hybrid_grid_features.m"	"Hybrid direct-hit and bracket detection"	true
"select_verified_hybrid_candidate.m"	"Hybrid verified-candidate tie-break"	true
"benchmark_shinbrot_search_methods.m"	"Single-source repeated timing benchmark"	true
"benchmark_shinbrot_source_ensemble.m"	"60-source deterministic ensemble benchmark"	true
"next_valid_section_crossing_noisy_rk4.m"	"Noisy fixed-step accepted crossing"	true
"rk4_lorenz_step.m"	"Fourth-order Runge-Kutta step"	true
"run_noise_sweep.m"	"Bisection-only noise sweep"	true
"run_noisy_targeting_trial.m"	"One noisy retargeting trial"	true
"build_bisection_noise_config.m"	"Noise configuration signature"	true
"load_valid_bisection_noise_results.m"	"Validated completed noise-result loader"	true
"plot_shinbrot_source_ensemble.m"	"Ensemble figures"	true
"plot_noise_results.m"	"Noise figures"	true

```
finalSummary = table( ...
  target.X, target.tolerance, sourceIndex, sourceX, ...
  bisectionSearch.selectedP, hybridSearch.selectedP, ...
  bisectionSearch.horizon, hybridSearch.horizon, ...
  bisectionReplay.targetError, hybridReplay.targetError, ...
  uncontrolled.numCrossings, ...
  'VariableNames', {'TargetX', 'TargetTolerance', 'SourceIndex', ...
    'SourceX', 'BisectionP', 'HybridP', 'BisectionHorizon', ...
    'HybridHorizon', 'BisectionReplayError', 'HybridReplayError', ...
    'UncontrolledCrossingsToHit'});
disp(finalSummary);
```

TargetX	TargetTolerance	SourceIndex	SourceX	BisectionP	HybridP	BisectionHorizon	HybridHorizon	BisectionReplayError	HybridReplayError	UncontrolledCrossingsToHit
13.729	0.008	150	12.807	0.086546	0.019375	3	3	4.0648e-09	0.00056305	266

```
fprintf('Shinbrot targeting walkthrough completed successfully.\n');
```

Shinbrot targeting walkthrough completed successfully.

Local Helper Functions

```
function projectPath = walkthrough_project_path()
```

```

activeFile = '';
try
    activeFile = matlab.desktop.editor.getActiveFilename;
catch
end
if ~isempty(activeFile) && isfile(activeFile)
    projectPath = fileparts(activeFile);
else
    projectPath = fileparts(mfilename('fullpath'));
    if isempty(projectPath) || ~isfolder(projectPath)
        projectPath = pwd;
    end
end
if ~isfile(fullfile(projectPath, 'lorenz_rhs.m'))
    projectPath = pwd;
end
if ~isfile(fullfile(projectPath, 'lorenz_rhs.m'))
    error('Could not locate the Shinbrot targeting project folder.');
```

```

end
end

function walkthrough_plot_lorenz_section(mapData, sourceState, target, params)
figure('Name', 'Lorenz attractor, accepted section, source and target', ...
    'Color', 'w');
plot3(mapData.X(:, 1), mapData.X(:, 2), mapData.X(:, 3), ...
    'Color', [0.08, 0.30, 0.68], 'LineWidth', 0.55);
hold on;
walkthrough_draw_section_half_plane(mapData.X, params);
scatter3(mapData.states(1, :), mapData.states(2, :), ...
    mapData.states(3, :), 10, [0.15, 0.45, 0.80], 'filled', ...
    'MarkerFaceAlpha', 0.25, 'DisplayName', 'Accepted crossings');
plot3(sourceState(1), sourceState(2), sourceState(3), 'ko', ...
    'MarkerFaceColor', 'y', 'MarkerSize', 8, 'DisplayName', 'Source');
plot3(target.state(1), target.state(2), target.state(3), 'rp', ...
    'MarkerFaceColor', 'r', 'MarkerSize', 13, 'DisplayName', 'Target');
grid on; box on; view(38, 24);
xlabel('x'); ylabel('y'); zlabel('z');
title(sprintf('Lorenz attractor and accepted section z = %.3f, x > %.1f', ...
    params.zSection, params.xSectionMin));
legend('Location', 'best');
hold off;
end

function walkthrough_draw_section_half_plane(X, params)
xLimits = [params.xSectionMin, max(X(:, 1))];
yLimits = [min(X(:, 2)), max(X(:, 2))];
[xPlane, yPlane] = meshgrid(xLimits, yLimits);
zPlane = params.zSection * ones(size(xPlane));
surf(xPlane, yPlane, zPlane, 'FaceColor', [0.95, 0.68, 0.10], ...
    'FaceAlpha', 0.22, 'EdgeColor', 'none', ...
    'DisplayName', 'Accepted half-plane');
end

function walkthrough_plot_return_map(mapData, sourceState, target)
figure('Name', 'Scalar Poincare return map', 'Color', 'w');
plot(mapData.xN, mapData.xNext, '.', 'Color', [0.18, 0.43, 0.72], ...
    'MarkerSize', 6, 'DisplayName', 'Return-map samples');
hold on;
mapLimits = [min([mapData.xN, mapData.xNext]), ...
    max([mapData.xN, mapData.xNext])];
plot(mapLimits, mapLimits, 'k--', 'LineWidth', 1, ...
    'DisplayName', 'Identity line');
plot(sourceState(1), target.x, 'ko', 'MarkerFaceColor', 'y', ...
    'MarkerSize', 8, 'DisplayName', 'Source scalar coordinate');
plot(target.x, target.x, 'rp', 'MarkerFaceColor', 'r', ...
    'MarkerSize', 13, 'DisplayName', 'Target fixed point');
grid on; box on; axis equal;
xlim(mapLimits); ylim(mapLimits);
xlabel('X_n');
ylabel('X_{n+1}');
title('Nominal scalar return map, pControl = 0');
legend('Location', 'best');
hold off;
end

function replay = walkthrough_replay_parameter(sourceState, target, params, search)
replay.hit = false;
replay.numCrossings = Inf;

```

```

replay.physicalTime = Inf;
replay.targetError = Inf;
replay.failed = false;
replay.t = [];
replay.X = [];
replay.states = [];
replay.xSequence = sourceState(1);
replay.replayTime = NaN;
if ~search.found || ~isfinite(search.selectedP) || ~isfinite(search.horizon)
    replay.failed = true;
    return;
end
currentState = sourceState(:);
currentTime = 0;
tFull = [];
XFull = [];
states = NaN(3, search.horizon);
for crossing = 1:search.horizon
    segment = next_valid_section_crossing( ...
        currentState, params, search.selectedP);
    if ~segment.success
        replay.failed = true;
        replay.numCrossings = crossing - 1;
        return;
    end
    [tFull, XFull] = walkthrough_append_segment( ...
        tFull, XFull, currentTime, segment);
    currentTime = currentTime + segment.eventTime;
    currentState = segment.eventState;
    states(:, crossing) = currentState;
end
replay.numCrossings = search.horizon;
replay.physicalTime = currentTime;
replay.targetError = abs(currentState(1) - target.x);
replay.hit = replay.targetError <= target.tolerance;
replay.t = tFull;
replay.X = XFull;
replay.states = states;
replay.xSequence = [sourceState(1), states(1, :)];
end

function [tFull, XFull] = walkthrough_append_segment(tFull, XFull, currentTime, segment)
tSegment = currentTime + segment.t;
if isempty(tFull)
    tFull = tSegment;
    XFull = segment.X;
else
    tFull = [tFull; tSegment(2:end)];
    XFull = [XFull; segment.X(2:end, :)];
end
end

function walkthrough_assert_valid_result(search, replay, target, control, label)
%WALKTHROUGH_ASSERT_VALID_RESULT Assert the properties of a valid result.
%
% A returned perturbation is acceptable when the search reports success,
% the horizon is finite and within the declared cap, the perturbation lies
% inside the admissible interval, the independent replay from the
% three-dimensional source state enters the target neighbourhood, and the
% search-reported error agrees with the replayed error.
%
% No particular value of p or of the final error is required. Different
% implementations of the search may return different admissible
% perturbations at the same horizon, and that is a property of the problem
% rather than an error.

assert(search.found, '%s: the search did not report a verified candidate.', label);
assert(isfinite(search.selectedP), '%s: selectedP is not finite.', label);
assert(abs(search.selectedP) <= control.deltaP + 1e-12, ...
    '%s: |p| = %.6g exceeds the admissible bound %.6g.', ...
    label, abs(search.selectedP), control.deltaP);
assert(isfinite(search.horizon) && search.horizon >= 1 && ...
    search.horizon <= control.maxSearchCrossings, ...
    '%s: horizon %g is outside 1..%d.', ...
    label, search.horizon, control.maxSearchCrossings);
assert(search.finalTargetError <= target.tolerance, ...
    '%s: search error %.6g exceeds the tolerance %.6g.', ...
    label, search.finalTargetError, target.tolerance);

```

```

assert(~replay.failed, '%s: the independent replay did not complete.', label);
assert(replay.hit, ...
    '%s: the independent replay did not enter the target interval (error %.6g).', ...
    label, replay.targetError);
assert(replay.numCrossings == search.horizon, ...
    '%s: the replay used %g crossings against a reported horizon of %g.', ...
    label, replay.numCrossings, search.horizon);

agreement = abs(search.finalTargetError - replay.targetError);
assert(agreement <= 0.01 * target.tolerance, ...
    '%s: search and replay errors differ by %.6g, more than 1%% of the tolerance.', ...
    label, agreement);

fprintf('%s: verified. p = %.10g, n = %d, replay error = %.6g, search-replay divergence = %.3g\n', ...
    label, search.selectedP, search.horizon, replay.targetError, agreement);

if agreement == 0
    fprintf([' Note: the divergence is exactly zero, so the replay repeated the search''s\n', ...
        ' own propagation rather than recomputing it independently. The search and the\n', ...
        ' replay reach the section through the same function, so agreement is guaranteed\n', ...
        ' and is not evidence that the returned perturbation is correct. The perturbation\n', ...
        ' is not thereby wrong; this particular check simply carries no weight for it.\n']);
end
end

function tbl = walkthrough_bisection_diagnostics(search, replay)
tbl = table(string(search.method), search.found, replay.hit, ...
    search.selectedP, abs(search.selectedP), search.horizon, ...
    search.finalTargetError, replay.targetError, search.searchTime, ...
    replay.replayTime, search.parameterEvaluations, ...
    search.crossingPropagations, search.totalBisectionIterations, ...
    search.crossingSignatureEvaluations, search.discontinuitiesDetected, ...
    search.discontinuityReductions, search.continuousIntervalsExamined, ...
    search.continuousTargetBracketFound, search.bracketLow, ...
    search.bracketHigh, string(search.selectionMethod), ...
    'VariableNames', {'Method', 'SearchFound', 'ReplayHit', ...
    'SelectedP', 'AbsP', 'Horizon', 'SearchTargetError', ...
    'ReplayTargetError', 'SearchTime', 'ReplayTime', ...
    'ParameterEvaluations', 'CrossingPropagations', ...
    'TotalBisectionIterations', 'CrossingSignatureEvaluations', ...
    'DiscontinuitiesDetected', 'DiscontinuityReductions', ...
    'ContinuousIntervalsExamined', 'ContinuousTargetBracketFound', ...
    'BracketLow', 'BracketHigh', 'SelectionMethod'});
end

function tbl = walkthrough_hybrid_diagnostics(search, replay)
tbl = table(string(search.method), search.found, replay.hit, ...
    search.selectedP, abs(search.selectedP), search.horizon, ...
    search.finalTargetError, replay.targetError, search.searchTime, ...
    replay.replayTime, search.parameterEvaluations, ...
    search.crossingPropagations, search.gridEvaluations, ...
    search.finalBracketBisectionIterations, ...
    search.totalBisectionIterations, search.bracketLow, ...
    search.bracketHigh, string(search.selectionMethod), ...
    'VariableNames', {'Method', 'SearchFound', 'ReplayHit', ...
    'SelectedP', 'AbsP', 'Horizon', 'SearchTargetError', ...
    'ReplayTargetError', 'SearchTime', 'ReplayTime', ...
    'ParameterEvaluations', 'CrossingPropagations', ...
    'GridEvaluations', 'FinalBracketBisectionIterations', ...
    'TotalBisectionIterations', 'BracketLow', 'BracketHigh', ...
    'SelectionMethod'});
end

function tbl = walkthrough_hybrid_feature_table(search, target)
horizons = (1:numel(search.directGridHitsByHorizon)).';
DirectGridHits = search.directGridHitsByHorizon(:);
AdjacentBrackets = search.adjacentBracketsByHorizon(:);
VerifiedCandidates = search.verifiedCandidatesByHorizon(:);
MinimumGridDistance = min(search.distanceByHorizon, [], 2, 'omitnan');
tbl = table(horizons, DirectGridHits, AdjacentBrackets, ...
    VerifiedCandidates, MinimumGridDistance, ...
    'VariableNames', {'Horizon', 'DirectGridHits', ...
    'AdjacentBrackets', 'VerifiedCandidates', 'MinimumGridDistance'});
tbl.TargetTolerance = target.tolerance * ones(height(tbl), 1);
end

function tbl = walkthrough_principal_comparison_table(bis, bisReplay, hyb, hybReplay)

```

```

Method = ["Shinbrot bisection"; "Grid-first hybrid"];
Success = [bis.found; hyb.found];
ReplayHit = [bisReplay.hit; hybReplay.hit];
SelectedControl = [bis.selectedP; hyb.selectedP];
AbsControl = abs(SelectedControl);
Horizon = [bis.horizon; hyb.horizon];
FinalTargetError = [bis.finalTargetError; hyb.finalTargetError];
ReplayTargetError = [bisReplay.targetError; hybReplay.targetError];
SearchTime = [bis.searchTime; hyb.searchTime];
ReplayTime = [bisReplay.replayTime; hybReplay.replayTime];
TotalComputationalTime = SearchTime + ReplayTime;
SelectionMethod = [string(bis.selectionMethod); string(hyb.selectionMethod)];
tbl = table(Method, Success, ReplayHit, SelectedControl, AbsControl, ...
    Horizon, FinalTargetError, ReplayTargetError, SearchTime, ...
    ReplayTime, TotalComputationalTime, SelectionMethod);
end

function walkthrough_plot_section_sequences(uncontrolled, bisReplay, hybReplay, target)
figure('Name', 'Controlled and uncontrolled section sequences', 'Color', 'w');
tiledlayout(1, 3, 'Padding', 'compact', 'TileSpacing', 'compact');
walkthrough_plot_one_sequence(uncontrolled.xSequence, target, ...
    'Uncontrolled recurrence', [0.08, 0.35, 0.75]);
walkthrough_plot_one_sequence(bisReplay.xSequence, target, ...
    'Shinbrot bisection replay', [0.12, 0.31, 0.47]);
walkthrough_plot_one_sequence(hybReplay.xSequence, target, ...
    'Grid-first hybrid replay', [0.84, 0.37, 0.00]);
end

function walkthrough_plot_one_sequence(xSequence, target, titleText, color)
nexttile;
n = 0:(numel(xSequence) - 1);
plot(n, xSequence, '-o', 'Color', color, 'MarkerFaceColor', color, ...
    'MarkerSize', 4, 'LineWidth', 1.1);
hold on;
xPatch = [0, max(n(end), 1), max(n(end), 1), 0];
yPatch = target.x + target.tolerance * [-1, -1, 1, 1];
patch(xPatch, yPatch, [0.88, 0.12, 0.12], 'FaceAlpha', 0.10, ...
    'EdgeColor', [0.88, 0.12, 0.12], 'LineWidth', 1);
ylines(target.x, 'r:', 'X_t', 'LineWidth', 1);
grid on; box on;
xlabel('Accepted section-crossing number');
ylabel('X_n');
title(titleText);
hold off;
end

function walkthrough_plot_controlled_phase_space(mapData, uncontrolled, ...
    bisReplay, hybReplay, sourceState, target, params)
figure('Name', 'Phase-space controlled and uncontrolled replays', ...
    'Color', 'w');
plot3(mapData.X(:, 1), mapData.X(:, 2), mapData.X(:, 3), ...
    'Color', [0.82, 0.82, 0.82], 'LineWidth', 0.35, ...
    'DisplayName', 'Nominal attractor');
hold on;
if ~isempty(uncontrolled.X)
    plot3(uncontrolled.X(:, 1), uncontrolled.X(:, 2), ...
        uncontrolled.X(:, 3), 'Color', [0.08, 0.35, 0.75], ...
        'LineWidth', 0.8, 'DisplayName', 'Uncontrolled');
end
if ~isempty(bisReplay.X)
    plot3(bisReplay.X(:, 1), bisReplay.X(:, 2), bisReplay.X(:, 3), ...
        'Color', [0.12, 0.31, 0.47], 'LineWidth', 1.4, ...
        'DisplayName', 'Shinbrot bisection');
end
if ~isempty(hybReplay.X)
    plot3(hybReplay.X(:, 1), hybReplay.X(:, 2), hybReplay.X(:, 3), ...
        'Color', [0.84, 0.37, 0.00], 'LineWidth', 1.4, ...
        'DisplayName', 'Grid-first hybrid');
end
walkthrough_draw_section_half_plane(mapData.X, params);
plot3(sourceState(1), sourceState(2), sourceState(3), 'ko', ...
    'MarkerFaceColor', 'y', 'MarkerSize', 8, 'DisplayName', 'Source');
plot3(target.state(1), target.state(2), target.state(3), 'rp', ...
    'MarkerFaceColor', 'r', 'MarkerSize', 13, 'DisplayName', 'Target');
grid on; box on; view(38, 24);
xlabel('x'); ylabel('y'); zlabel('z');
title('Independent controlled replays from the same source');
legend('Location', 'best');

```

```

hold off;
end

function walkthrough_plot_search_behaviour(bis, hyb, target, control)
figure('Name', 'Deterministic search behaviour', 'Color', 'w');
tiledlayout(1, 2, 'Padding', 'compact', 'TileSpacing', 'compact');

nexttile;
bar(categorical({'Discontinuities', 'Reductions', 'Continuous intervals', ...
    'Signature evaluations'}), ...
    [bis.discontinuitiesDetected, bis.discontinuityReductions, ...
    bis.continuousIntervalsExamined, bis.crossingSignatureEvaluations]);
hold on;
if isfinite(bis.bracketLow)
    subtitle(sprintf('Selected p = %.8g, bracket [%.8g, %.8g]', ...
        bis.selectedP, bis.bracketLow, bis.bracketHigh));
end
ylabel('count');
title('Shinbrot bisection diagnostics');
grid on; box on;

nexttile;
hasGridCurve = any(isfinite(hyb.gridDistance));
if hasGridCurve
    semilogy(hyb.pGrid, max(hyb.gridDistance, eps), '-o', ...
        'Color', [0.84, 0.37, 0.00], 'MarkerSize', 3, ...
        'MarkerFaceColor', [0.84, 0.37, 0.00], ...
        'DisplayName', 'Grid residual');
    hold on;
end
yline(target.tolerance, 'r--', 'Target tolerance', 'LineWidth', 1.2);
if isfinite(hyb.bracketLow)
    xline(hyb.bracketLow, ':', 'Bracket low', 'LineWidth', 1.1);
end
if isfinite(hyb.bracketHigh)
    xline(hyb.bracketHigh, ':', 'Bracket high', 'LineWidth', 1.1);
end
if hyb.found
    xline(hyb.selectedP, 'k--', 'Selected pControl', 'LineWidth', 1.2);
end
xlim([-control.deltaP, control.deltaP]);
xlabel('pControl');
ylabel('|X_n - X_t|');
title(sprintf('Hybrid grid residual, horizon n = %d', hyb.plotHorizon));
grid on; box on;
legend('Location', 'best');
end

function rhsRms = walkthrough_compute_rhs_rms(mapData, params, control, transientCrossingIndex)
transientTime = mapData.eventTimes(transientCrossingIndex);
keep = mapData.t >= transientTime;
t = mapData.t(keep);
X = mapData.X(keep, :);
duration = t(end) - t(1);
F = NaN(size(X));
for i = 1:size(X, 1)
    F(i, :) = lorenz_rhs(t(i), X(i, :).', params, 0).';
end
rhsRms.rmsFx = sqrt(trapz(t, F(:, 1).^2) / duration);
rhsRms.rmsFy = sqrt(trapz(t, F(:, 2).^2) / duration);
rhsRms.rmsFz = sqrt(trapz(t, F(:, 3).^2) / duration);
rhsRms.transientCrossingIndex = transientCrossingIndex;
rhsRms.transientTime = transientTime;
rhsRms.averagingDuration = duration;
rhsRms.maxParameterEffect = control.deltaP;
rhsRms.parameterToRmsFyRatio = control.deltaP / rhsRms.rmsFy;
rhsRms.paperApproximateRmsFy = 100;
rhsRms.rmsFyDifferenceFromPaperApproximation = ...
    rhsRms.rmsFy - rhsRms.paperApproximateRmsFy;
end

function valid = walkthrough_is_valid_source_ensemble(saved, target, control)
required = {'sourceIndices', 'methodResultsTable', ...
    'pairedResultsTable', 'summary', 'methodSummary', ...
    'studySettings', 'target', 'control'};
valid = all(isfield(saved, required));
if ~valid
    return;
end

```

```

end
methods = saved.methodResultsTable.Method;
methodVariables = saved.methodResultsTable.Properties.VariableNames;
pairedVariables = saved.pairedResultsTable.Properties.VariableNames;
valid = istable(saved.methodResultsTable) && ...
    istable(saved.pairedResultsTable) && istable(saved.summary) && ...
    istable(saved.methodSummary) && ...
    numel(saved.sourceIndices) == 60 && ...
    numel(unique(saved.sourceIndices)) == 60 && ...
    all(saved.sourceIndices >= 100 & saved.sourceIndices <= 300) && ...
    height(saved.methodResultsTable) == 120 && ...
    height(saved.pairedResultsTable) == 60 && ...
    sum(strcmp(methods, verified_bisection_identififer())) == 60 && ...
    sum(strcmp(methods, 'hybridGridBisection')) == 60 && ...
    all(saved.methodResultsTable.GridEvaluations(strcmp(methods, ...
        verified_bisection_identififer())) == 0) && ...
    all(ismember({'FailureCode', 'FailureReason'}, methodVariables)) && ...
    all(ismember({'ShinbrotFinalTargetError', ...
        'HybridFinalTargetError', 'ShinbrotReplayTargetError', ...
        'HybridReplayTargetError'}, pairedVariables)) && ...
    saved.studySettings.numSources == 60 && ...
    saved.studySettings.sourceSeparation == 1.5 && ...
    isequal(saved.studySettings.candidateIndices(:).', 100:300) && ...
    saved.control.maxSearchCrossings == control.maxSearchCrossings && ...
    saved.control.deltaP == control.deltaP && ...
    saved.target.x == target.x && ...
    saved.target.tolerance == target.tolerance;
end

function walkthrough_plot_ensemble_distributions(sourceEnsemble)
paired = sourceEnsemble.pairedResultsTable;
figure('Name', 'Ensemble error and search-time distributions', 'Color', 'w');
tiledlayout(1, 2, 'Padding', 'compact', 'TileSpacing', 'compact');

nexttile;
histogram(paired.ShinbrotReplayTargetError(paired.ShinbrotReplayHit), ...
    'FaceColor', [0.12, 0.31, 0.47], 'FaceAlpha', 0.65, ...
    'DisplayName', 'Shinbrot');
hold on;
histogram(paired.HybridReplayTargetError(paired.HybridReplayHit), ...
    'FaceColor', [0.84, 0.37, 0.00], 'FaceAlpha', 0.55, ...
    'DisplayName', 'Hybrid');
xlabel('Replay target error');
ylabel('source count');
title('Verified replay target errors');
legend('Location', 'best');
grid on; box on;

nexttile;
methods = sourceEnsemble.methodResultsTable.Method;
shinbrotTimes = sourceEnsemble.methodResultsTable.SearchTime( ...
    strcmp(methods, verified_bisection_identififer()));
hybridTimes = sourceEnsemble.methodResultsTable.SearchTime( ...
    strcmp(methods, 'hybridGridBisection'));
boxchart(categorical([repmat("Shinbrot", numel(shinbrotTimes), 1); ...
    repmat("Hybrid", numel(hybridTimes), 1)]), ...
    [shinbrotTimes; hybridTimes]);
ylabel('Search time');
title('Single-run ensemble search times');
grid on; box on;
end

function sweep = walkthrough_noise_sweep_settings(verifiedIdentifier, sourceIndex, isSmokeTest)
if isSmokeTest
    sweep.sigmaNoiseValues = [0, 0.05];
    sweep.numTrials = 1;
    sweep.baseSeed = 191000;
    sweep.outputMatFile = 'noise_smoke_bisection_section_retargnet.mat';
    sweep.checkpointMatFile = ...
        'noise_smoke_bisection_section_retargnet_checkpoint.mat';
    sweep.summaryCsvFile = 'noise_smoke_summary_bisection_section_retargnet.csv';
    sweep.trialsCsvFile = 'noise_smoke_trials_bisection_section_retargnet.csv';
    sweep.resumeFromCheckpoint = false;
else
    sweep.sigmaNoiseValues = ...
        [0, 0.01, 0.03, 0.05, 0.075, 0.1, 0.15, 0.2, 0.3, 0.5, 1.0];
    sweep.numTrials = 10;
    sweep.baseSeed = 91000;

```

```

    sweep.outputMatFile = 'noise_results_bisection_section_retargnet.mat';
    sweep.checkpointMatFile = ...
        'noise_results_bisection_section_retargnet_checkpoint.mat';
    sweep.summaryCsvFile = 'noise_summary_bisection_section_retargnet.csv';
    sweep.trialsCsvFile = 'noise_trials_bisection_section_retargnet.csv';
    sweep.resumeFromCheckpoint = true;
end

sweep.modes = {'sectionRetargetMode'};
sweep.searchMethod = verifiedIdentifier;
sweep.sourceIndex = sourceIndex;
sweep.isSmokeTest = isSmokeTest;
end

function walkthrough_plot_noise_summary(noiseResults)
summary = noiseResults.summaryTable;
figure('Name', 'Noise summary compact view', 'Color', 'w');
tiledlayout(1, 2, 'Padding', 'compact', 'TileSpacing', 'compact');

nexttile;
plot(summary.SigmaNoise, summary.SuccessFraction, 'o-', ...
    'LineWidth', 1.5, 'MarkerFaceColor', [0.12, 0.31, 0.47], ...
    'Color', [0.12, 0.31, 0.47]);
ylim([-0.05, 1.05]);
xlabel('sigma_noise');
ylabel('success fraction');
title('Noise-sweep success fraction');
grid on; box on;

nexttile;
plot(summary.SigmaNoise, summary.MeanRetargetingEvents, 's-', ...
    'LineWidth', 1.5, 'MarkerFaceColor', [0.84, 0.37, 0.00], ...
    'Color', [0.84, 0.37, 0.00]);
xlabel('sigma_noise');
ylabel('mean retargeting events');
title('Accepted-section retargeting effort');
grid on; box on;
end

function files = walkthrough_supporting_files_table(projectPath)
names = [
    "lorenz_rhs.m"
    "poincare_event.m"
    "next_valid_section_crossing.m"
    "generate_return_map.m"
    "uncontrolled_hitting_time_1d.m"
    "search_parameter_to_target.m"
    "run_shinbrot_discontinuity_aware_bisection.m"
    "evaluate_shinbrot_crossing_signature.m"
    "find_hybrid_grid_features.m"
    "select_verified_hybrid_candidate.m"
    "benchmark_shinbrot_search_methods.m"
    "benchmark_shinbrot_source_ensemble.m"
    "next_valid_section_crossing_noisy_rk4.m"
    "rk4_lorenz_step.m"
    "run_noise_sweep.m"
    "run_noisy_targeting_trial.m"
    "build_bisection_noise_config.m"
    "load_valid_bisection_noise_results.m"
    "plot_shinbrot_source_ensemble.m"
    "plot_noise_results.m"
];
roles = [
    "Lorenz vector field"
    "Poincare plane event"
    "Accepted deterministic section crossing"
    "Nominal scalar return-map data"
    "Uncontrolled recurrence baseline"
    "Dispatcher for both deterministic searches"
    "Verified Shinbrot bisection search"
    "Crossing-order signature diagnostic"
    "Hybrid direct-hit and bracket detection"
    "Hybrid verified-candidate tie-break"
    "Single-source repeated timing benchmark"
    "60-source deterministic ensemble benchmark"
    "Noisy fixed-step accepted crossing"
    "Fourth-order Runge-Kutta step"
    "Bisection-only noise sweep"
    "One noisy retargeting trial"

```

```
"Noise configuration signature"
"Validated completed noise-result loader"
"Ensemble figures"
"Noise figures"
];
exists = false(numel(names), 1);
for i = 1:numel(names)
    exists(i) = isfile(fullfile(projectPath, names(i)));
end
files = table(names, roles, exists, ...
    'VariableNames', {'FileName', 'Role', 'Exists'});
end
```